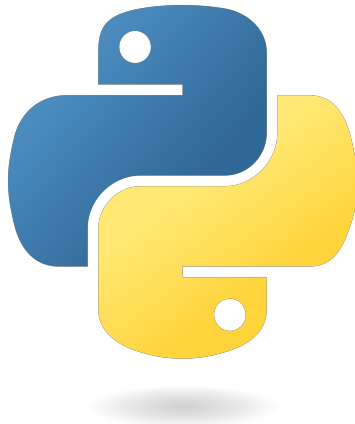


PYTHON IMPLEMENTATION



A key and recurring theme in finance is the pervasive uncertainty surrounding the performance of almost every tradable asset, which can manifest itself as increased volatility in prices, interest rates, and exchange rates. This volatility naturally poses substantial risks to portfolios and financial positions, prompting the need for robust strategies to mitigate potential losses. One widely discussed approach is diversification, a tactic that implicitly offers a “free lunch,” as demonstrated in Chapter 1, by dispersing individual asset risk across a wider set of holdings. However, diversification primarily shields against idiosyncratic or unsystematic risk, and when the entire market changes downward, it proves insufficient to protect portfolios from broad market declines.

Among the various techniques for managing risk, hedging stands out as a particularly effective method to minimize unwanted exposures in a portfolio. In essence, it involves taking positions, often through financial instruments such as options, futures, or swaps, that can offset potential losses arising from adverse market movements. This approach may involve letting go of some potential upside profit, similar to paying an “insurance premium,” to guard against severe downside outcomes. By carefully calibrating these hedging strategies, we can aim for a steadier and more predictable financial outcome, one that remains neutral to market fluctuations. In practical terms, this resembles purchasing insurance against the damaging effects of sudden price changes, interest rate changes, or currency fluctuations, thus providing an additional layer of protection and contributing to a more balanced risk-return profile.

Hedging can be executed with a wide range of financial instruments, commonly via financial derivatives such as futures and options. By definition, derivatives derive their value from underlying assets, such as stocks, bonds, commodities, or currencies, offering a degree of risk management without requiring direct ownership of the underlying. This inherent flexibility makes them especially attractive for hedging, where one pays a relatively small premium to shield against the potential for severe losses. In practice, this means that investors can strategically construct

positions that offset adverse moves in the market, thereby mitigating portfolio volatility and preserving more stable returns.

In this chapter, we examine two prominent derivatives, futures and options, that are central to hedging strategies. Futures contracts, in particular, are standardized agreements that are used to buy or sell an underlying asset at a predetermined price on a specific future date. Because the price is locked in at the time the contract is created, subsequent market fluctuations no longer affect the agreed-upon transaction value, ensuring a degree of certainty in otherwise volatile conditions. This feature is particularly advantageous for commodity market participants such as farmers, who may use futures to lock in the selling price of their crops, and manufacturers, who can stabilize the cost of raw materials purchased at later dates. More generally, futures contracts provide a practical means of protecting against price fluctuations in a variety of markets, including commodities, interest rates, and other financial products. Traded on regulated exchanges, these contracts benefit from high liquidity, making them accessible for many hedging scenarios. In addition, their standardized terms, governed by centralized exchanges, enhance transparency and reduce counterparty risk through margin requirements and clearing mechanisms, thus simplifying the implementation of hedging strategies.

When holding option contracts, we gain the right, but not the obligation, to buy (in the case of a call option) or sell (in the case of a put option) an underlying asset at a specified strike price within a predetermined time frame. This fundamental distinction from futures, where both parties are obligated to transact, gives option holders the flexibility to simply let the contract expire if the market outlook becomes unfavorable. Consequently, investors can limit their downside while retaining the ability to capture upside gains should prices move in a beneficial direction. This asymmetry in risk and return helps protect against adverse market moves and preserves the potential for profit, making options one of the most popular and versatile hedging instruments in financial markets.

Options, as financial instruments, also experience price fluctuations driven by changes in the underlying asset, time to expiration, and implied volatility. A well-known hedging technique to manage these fluctuations is delta hedging, which involves quantifying how an option's price responds to small changes in the price of the underlying asset, such as stock. This sensitivity is called the delta of the option (Δ). Achieving a "delta neutral" position means compensating for this sensitivity by adjusting the number of underlying assets or related derivatives held, thus reducing exposure to the movements of the underlying price. Because market conditions can change rapidly, maintaining a delta-neutral stance requires continuous monitoring and frequent rebalancing of positions, making this a dynamic and complex strategy to implement. Nonetheless, the systematic approach of delta hedging presents significant opportunities to manage risk effectively, as we will explore in the following sections.

In this chapter, we will explore hedging strategies that use futures and options, paying special attention to the principles and applications of delta hedging. We begin by dissecting the mechanics underpinning these derivatives—covering contract specifications, pricing, and market conventions—to establish a solid foundation.

Building on this groundwork, we then examine various hedging tactics and the theoretical underpinnings that make them effective.

Let us start with the futures contract.

5.1 Hedging with Futures Contracts

Futures contracts are widely used to mitigate exposure to price fluctuations in commodities, currencies, and other financial instruments. They are standardized legal agreements that require two parties to buy or sell a specified quantity of an underlying asset at a set price on a predetermined future date. Crucially, exchanges govern the terms, such as contract size, expiration date, and quality or delivery specifications, ensuring consistency between trades and supporting a liquid market. Although these contracts outline a physical delivery process, market participants aiming to hedge can opt to close or “roll over” their positions rather than receive or deliver the physical asset. In this way, futures serve as a versatile tool, allowing investors and commercial entities to lock in future prices without necessarily incurring the logistical burden of actual commodity exchange.

5.1.1 Hedging Mechanism Using Futures

The fundamental concept behind futures hedging is to establish a position in the futures market that is counter to one’s position in the spot (cash) market, helping to offset adverse price movements in the underlying asset. For example, if you already hold an asset and fear a drop in its price, you can sell (short) futures contracts, known as a short hedge, to protect against potential losses. Conversely, if you anticipate buying an asset in the future and worry about rising prices, you can buy (go long) futures contracts to lock in a predetermined price. This so-called long hedge ensures that you can secure the asset at a set cost, effectively insulating your budget from market volatility.

In both cases, movements in the futures market tend to counterbalance fluctuations in the spot market, reducing overall risk. Thus, these offsetting positions offer a practical and cost-effective means of stabilizing cash flows and mitigating the impact of unpredictable price changes. The net effect is to reduce exposure to unfavorable price movements in the underlying asset. However, note that some residual discrepancy, often referred to as “basis risk,” may persist because of differences between the price of the spot and futures.

To illustrate the hedging process, consider the following scenario. Let S_t denote the spot price of an asset at time t , and Q represent the total quantity of that asset either held (in a selling context) or required (in a buying context). The value of the spot position at time t , denoted by V_{spot} , can be expressed as

$$V_{\text{spot}} = Q \times S_t$$

which captures the portion of a portfolio that depends directly on the current market price of the asset. Next, let F_t be the futures price at time t , N the number of futures contracts (positive if taking a long position, negative if short), and q the contract size, which specifies how many units of the underlying are covered by each contract. The value of this futures position, V_{futures} , is thus given by

$$V_{\text{futures}} = N \times q \times F_t$$

By examining both V_{spot} and V_{futures} in tandem, we can see how changes in the spot price can be counterbalanced by gains or losses in the futures position, thus forming the core mechanism behind many hedging strategies.

When we enter into a futures contract at the initiation time ($t = 0$), the net value of our portfolio from both spot and futures positions (aside from any margin requirements) at any time t can be expressed as

$$V_{\text{total}} = V_{\text{spot}} + V_{\text{futures}} = Q \times S_t + N \times q \times F_t$$

The profit or loss (Π) from the hedged position during the hedging period, from $t = 0$ to $t = T$, can then be calculated by evaluating changes in both the spot and future positions. That is

$$\Pi = [V_{\text{spot}}(T) - V_{\text{spot}}(0)] + [V_{\text{futures}}(T) - V_{\text{futures}}(0)]$$

Since the initial value of the futures position is $V_{\text{futures}}(0) = N \times q \times F_0$, the change in the futures position value becomes

$$\Delta V_{\text{futures}} = V_{\text{futures}}(T) - V_{\text{futures}}(0) = N \times q \times (F_T - F_0)$$

Similarly, we can write the change in the spot position as

$$\Delta V_{\text{spot}} = S_T - S_0$$

Substituting this into Π gives

$$\Pi = \Delta V_{\text{spot}} + \Delta V_{\text{futures}} = Q \times (S_T - S_0) + N \times q \times (F_T - F_0)$$

At the contract's maturity, the futures price F_T converges to the spot price S_T , meaning $F_T = S_T$. Plugging this into Π simplifies to

$$\Pi = Q \times (S_T - S_0) + N \times q \times (S_T - F_0)$$

It is important to note that the sign of N , the number of futures contracts, gives an indication of the nature of the hedge. A negative N indicates a short position, which means that you have sold futures contracts, while a positive N indicates a long position, meaning you have bought futures contracts.

Let us consider the case of a short hedge, where we long the spot position and short the futures position as a hedge. Specifically, we hold Q units of the underlying asset and want protection against a potential drop in its price. To do so, we sell (short) futures contracts, where the number of contracts N is typically chosen so that $N \times q \approx Q$. As discussed above, during the hedging period, the value of the spot position changes by $Q (S_T - S_0)$, while the futures position changes by $N q (F_T - F_0)$. With $N < 0$ (due to short position in futures contract) and $N \times q = -Q$, these changes combine to

$$\Pi = Q (S_T - S_0) + N q (F_T - F_0) = Q (S_T - S_0) - Q (F_T - F_0).$$

At maturity, assume F_T converges to S_T (which is often the case based on the no arbitrage argument), so substituting $F_T = S_T$ eliminates the dependence on the final spot price S_T , yielding

$$\Pi = Q (F_0 - S_0).$$

This fixed payoff means that we have eliminated the exposure to the asset's price movements due to independence with S_T . Therefore, the short position in futures is considered a perfect hedge in the sense of removing price risk, although the final profit or loss Π can still be positive, negative, or zero depending on the initial difference $(F_0 - S_0)$.

5.1.2 Optimal Hedge Ratio

The effectiveness of a futures hedge largely hinges on how closely the movement in futures prices tracks the corresponding spot price of the asset. Although the idealized assumption $F_T = S_T$ implies that the futures price converges precisely to the terminal spot price, real-world factors, such as carrying costs, delivery specifications, and timing mismatches, can create a nonzero discrepancy known as basis risk at any time point in between $t = 0$ and $t = T$. As a result, the profit or loss of the hedged portfolio (Π_t) may be positive or negative at any time, rather than strictly zero. In practice, rather than enforcing $\Pi_t = 0$, a common objective is to minimize the variance σ_{Π}^2 of the portfolio's returns. One way to achieve this is to determine the optimal hedge ratio h^* , which specifies the proportion of the underlying position to hedge. Under the assumption that the returns of the spot and futures follow a joint normal distribution, the optimal ratio can be derived as

$$h^* = -\rho_{SF} \frac{\sigma_S}{\sigma_F},$$

where ρ_{SF} is the correlation coefficient between the changes in the price of the spot and futures, σ_S is the standard deviation of the changes in spot price, and σ_F is the

standard deviation of the changes in futures price. This formula highlights the key role of both correlation and relative volatility in designing an effective hedge.

To derive this formula, we analyze the variance of the hedged position and determine the value of the hedge ratio (while treating other variables as constant) that minimizes this variance. Recall that the hedged position consists of both the spot position and the futures position, and the change in the value of the hedged position Π over a short time interval can be expressed as follows:

$$\Pi = \Delta V_{\text{spot}} + \Delta V_{\text{futures}} = Q \times \Delta S + N \times q \times \Delta F$$

where Q is the quantity of the asset held or required, N is the number of futures contracts held (again, positive for a long position and negative for a short position), and q is the quantity of the asset per futures contract.

We define the hedge ratio h as a relative ratio:

$$h = \frac{N \times q}{Q}$$

This allows us to express N as

$$N = h \times \frac{Q}{q}$$

Substituting this into the expression for Π gives

$$\Pi = Q \times (\Delta S + h \times \Delta F)$$

The variance of the hedged position is then

$$\sigma_{\Pi}^2 = \text{Var}(\Pi) = Q^2 \times \text{Var}(\Delta S + h \times \Delta F)$$

Noting that the random variables are ΔS and ΔF , expanding the variance gives

$$\sigma_{\Pi}^2 = Q^2 \left[\text{Var}(\Delta S) + h^2 \text{Var}(\Delta F) + 2h \text{Cov}(\Delta S, \Delta F) \right]$$

Dividing both sides by Q^2 simplifies the equation to

$$\frac{\sigma_{\Pi}^2}{Q^2} = \sigma_S^2 + h^2 \sigma_F^2 + 2h \rho_{SF} \sigma_S \sigma_F$$

To find the hedge ratio that minimizes the variance σ_{Π}^2 , we can take the derivative of σ_{Π}^2 with respect to h (now we treat other variables as fixed) and set it to zero:

$$\frac{d\sigma_{\Pi}^2}{dh} = 2h\sigma_F^2 + 2\rho_{SF}\sigma_S\sigma_F = 0$$

Solving for h^* gives

$$h^* = -\rho_{SF} \frac{\sigma_S}{\sigma_F}$$

The negative sign indicates that the futures position should be opposite to the spot position to effectively minimize risk. This means that if we have a long position in the spot market, we should take a short position in futures contract and vice versa. However, since N is already defined as negative for a short position, the optimal hedge ratio can be expressed without the negative sign:

$$h^* = \rho_{SF} \times \frac{\sigma_S}{\sigma_F}$$

Thus, this expression determines the proportion of the exposure (percentage of total number of spot assets) that should be hedged to achieve minimal variance in the hedged position.

For example, assume $\sigma_S = 2$, $\sigma_F = 1.5$, and $\rho_{SF} = 0.5$. The optimal hedge ratio can be calculated using these values as follows:

$$h^* = 0.5 \times \frac{2}{1.5} \approx 0.6667$$

This result indicates that approximately 66.67% of the exposure should be hedged. Generally, a lower correlation coefficient leads to a lower hedge ratio because the effectiveness of the futures contract in hedging the spot position diminishes as the correlation between their price changes weakens.

Once the optimal hedge ratio is determined, the number of futures contracts N^* that is required to hedge the exposure can be calculated via

$$N^* = -h^* \times \frac{Q}{q}$$

The negative sign reflects taking a position opposite to the spot position. If, for instance, the exposure is $Q = 100,000$ units and each futures contract covers $q = 1,000$ units, then

$$N^* = -0.6667 \times \frac{100,000}{1,000} = -0.6667 \times 100 = -66.67$$

Since trading a fraction of a contract is not feasible, we typically round to the nearest whole number, resulting in selling 67 futures contracts to hedge the exposure.

Again, note that even with an optimal hedge ratio, perfect hedging is rarely achievable due to basis risk, which refers to the potential risk that futures prices do not move in sync with spot prices. This discrepancy can arise from differences

in the underlying asset, contract specifications, or timing mismatches. Furthermore, implementing a hedge comes with transaction costs that should be considered along with the benefits of hedging.

5.1.3 Scenario Analysis at Maturity

To illustrate how the hedge ratio and the correlation factor influence the effectiveness of a hedge and the terminal payoff, let us look at a scenario analysis. We will examine two possible outcomes at maturity: one in which the spot price decreases, leading to $S_T < S_0$, and another in which it increases, leading to $S_T > S_0$. This analysis will reveal how the hedged position responds in each scenario.

Consider the initial condition with a spot price of $S_0 = \$50$ per unit and futures price of $F_0 = \$50$ per unit (which may not necessarily be a realistic assumption due to other considerations on the cost of carry, such as financing costs, storage costs, convenience yields, and dividends). Assume that the quantity of the asset is $Q = 100,000$ units and that each futures contract covers $q = 1,000$ units. Also, assume that the optimal hedge ratio is $h^* = 0.4$. Using the optimal hedge ratio, we calculate the number of futures contracts needed as

$$N^* = -h^* \times \frac{Q}{q} = -0.4 \times \frac{100,000}{1,000} = -40 \text{ contracts}$$

The negative sign indicates a short position in futures contracts.

First, we analyze the situation where the spot price S_T drops to \$45 at maturity. The change in the spot market is calculated as

$$\Delta V_{\text{spot}} = Q \times (S_T - S_0) = 100,000 \times (\$45 - \$50) = -\$500,000$$

which represents a loss of \$500,000 due to the decrease in the spot price.

Next, we assume that the futures price at maturity is $F_T = \$43$, reflecting an imperfect correlation between the spot and futures prices (an example of basis risk where spot and futures prices do not move in perfect lockstep). The change in the futures price is

$$\Delta F = F_T - F_0 = \$43 - \$50 = -\$7$$

Thus, the gain from the futures position is then

$$\Delta V_{\text{futures}} = N^* \times q \times \Delta F = (-40) \times 1,000 \times (-\$7) = \$280,000$$

The total profit or loss Π from the hedged position is

$$\Pi = \Delta V_{\text{spot}} + \Delta V_{\text{futures}} = -\$500,000 + \$280,000 = -\$220,000$$

In this scenario, the loss in the spot market is partially offset by the gain in the futures market, leading to a net loss of \$220,000. This indicates that while the hedge has reduced the overall risk, it has not completely eliminated it. This is due to the imperfect correlation between the markets and the fact that only 40% of the exposure was hedged.

Let us look at another scenario where the spot price now rises to \$55 at maturity. The change (gain) in the spot market is now

$$\Delta V_{\text{spot}} = Q \times (S_T - S_0) = 100,000 \times (\$55 - \$50) = \$500,000$$

which represents a gain of \$500,000 due to the increase in the spot price. Now assuming the futures price at maturity is $F_T = \$54$, the change in the futures price is

$$\Delta F = F_T - F_0 = \$54 - \$50 = \$4$$

The loss from the futures position is then

$$\Delta V_{\text{futures}} = N^* \times q \times \Delta F = (-40) \times 1,000 \times \$4 = -\$160,000$$

The total profit or loss Π for the hedged position is

$$\Pi = \Delta V_{\text{spot}} + \Delta V_{\text{futures}} = \$500,000 - \$160,000 = \$340,000$$

In this scenario, the gain in the spot market is partially offset by the loss in the futures market, resulting in a net profit of \$340,000. This outcome shows that the hedged position limits potential gains, which is an expected result when taking up a hedging strategy.

In general, hedging reduces the downside risk by limiting the upside gain. It does not entirely eliminate the risk due to the imperfect correlation between the spot and futures prices. Specifically, only 40% of the exposure was hedged. Because the futures price does not move perfectly in line with the spot price, this leads to what is known as basis risk. Moreover, since only a portion of the exposure is hedged, gains and losses in the spot market are only partially offset by the futures position. This analysis emphasizes the importance of understanding the limitations of hedging strategies and the factors that influence their effectiveness.

5.1.4 Consideration of Basis Risk

The basis at any given time t is defined as the difference between the spot price and the futures price of an asset, which can be expressed as

$$\text{Basis}_t = S_t - F_t$$

At maturity ($t = T$), the basis is expected to be zero because the futures price F_T should converge to the spot price S_T ; otherwise, arbitrage activities can occur. This convergence is a fundamental principle in futures markets, ensuring that the futures contract and the underlying asset are priced consistently as the contract approaches expiration. However, in reality, the basis may not be exactly zero at maturity. This discrepancy arises because of an imperfect correlation between the spot and futures prices and various market inefficiencies. For example, it could be due to differences in the characteristics of the underlying asset, variations in the specifications of the contract, or the timing of the hedge being closed out before the futures contract matures. When the futures price does not move in sync with the spot price, the hedge fails to eliminate risk, leaving some residual exposure completely. This residual basis risk leads to $F_T \neq S_T$.

The effectiveness of a hedge can be assessed by comparing the variance of the hedged position to that of the unhedged position, where the variance is calculated based on the entire duration from contract initiation to maturity. This comparison is quantified by the reduction in variance, which is expressed via

$$\Delta\sigma^2 = \sigma_{\text{unhedged}}^2 - \sigma_{\text{hedged}}^2$$

Here, $\sigma_{\text{unhedged}}^2$ denotes the variance of the spot position alone, and we would expect it to be higher than σ_{hedged}^2 , which represents the variance of the combined positions in both the spot and futures markets. Ideally, a perfect hedge would eliminate all risk, resulting in $\sigma_{\text{hedged}}^2 = 0$ and thus $\Delta\sigma^2 = \sigma_{\text{unhedged}}^2$. However, due to basis risk, the reduction in variance $\Delta\sigma^2$ is typically less than the total variance of the spot position. This echoes our previous message that, while the hedge reduces the overall risk, it does not completely eliminate it.

Basis risk affects the effectiveness of a hedge because a futures contract may not completely offset the price changes of the underlying asset. As a result, even when using an optimal hedge ratio, some level of risk persists. Therefore, this residual risk needs to be recognized and managed, as it can influence the stability and predictability of the performance of the hedged position.

5.1.5 Implementing the Dynamic Hedging Strategy

In previous sections, we covered the fundamentals of hedging with futures contracts and the use of the optimal hedge ratio to minimize the variance of a hedged position. However, financial markets are dynamic, with volatilities, correlations, and other parameters changing over time. To adapt to these changes, we introduced dynamic hedging, which involves periodically adjusting the hedge position in response to evolving market conditions. In this section, we will illustrate the implementation of a dynamic futures hedging strategy using Python. We first use the Geometric Brownian Motion (GBM) process to simulate asset prices, then apply linear

regression to estimate/forecast the optimal hedge ratios. Finally, we dynamically adjust the hedge over time.

To create realistic simulations of price movements, we use GBM, a continuous-time stochastic process commonly applied in financial modeling. GBM effectively captures key characteristics of asset price dynamics, including the log-normal distribution of prices and the important fact that prices cannot fall below zero. The stochastic differential equation (SDE) for GBM representing the asset price S_t is

$$dS_t = \mu_S S_t dt + \sigma_S S_t dW_t^S$$

where μ_S is the expected return (drift) of the asset, σ_S is the volatility of the asset, and dW_t^S is the increment of a Wiener process (standard Brownian motion).

Similarly, we can model the SDE of the futures price F_t as

$$dF_t = \mu_F F_t dt + \sigma_F F_t dW_t^F$$

where μ_F and σ_F are the drift and volatility of the futures price, and dW_t^F is another Wiener process. To model the correlation between the asset and futures prices, we assume the dependence structure:

$$dW_t^S dW_t^F = \rho_{SF} dt$$

where ρ_{SF} is the correlation coefficient between changes in asset price S_t and futures F_t . For numerical simulation, we can discretize these SDEs to obtain the closed-form pricing formula for the next period:

$$S_{t+\Delta t} = S_t \times \exp \left(\left(\mu_S - \frac{\sigma_S^2}{2} \right) \Delta t + \sigma_S \epsilon_S \sqrt{\Delta t} \right)$$

$$F_{t+\Delta t} = F_t \times \exp \left(\left(\mu_F - \frac{\sigma_F^2}{2} \right) \Delta t + \sigma_F \epsilon_F \sqrt{\Delta t} \right)$$

where ϵ_S and ϵ_F are standard normal random variables with correlation ρ_{SF} . Note that all asset prices are positive (or zero, theoretically), and this constraint is satisfied in these pricing formulas.

As time passes and the market moves, the optional hedge ratio h^* also changes. To dynamically estimate h^* , we can perform a linear regression to model the changes in S_t as a function of changes in F_t based on the simulated changes at each time step. The starting point is to consider a spot position in an asset with price changes ΔS_t and a futures position with price changes ΔF_t . If we define the hedge ratio h as the number of futures contracts (scaled appropriately) held per unit of the underlying asset, then the total change in the value of the hedged portfolio can be written as $\Pi = \Delta S_t - h \Delta F_t$ when the underlying is long and the futures position is short. As introduced earlier, to determine h that minimizes risk,

we can seek to minimize the variance of Π , denoted $\text{Var}(\Pi)$. By expanding this variance and treating ΔS_t and ΔF_t as jointly normal random variables, it follows that $\text{Var}(\Pi) = \text{Var}(\Delta S_t - h \Delta F_t) = \text{Var}(\Delta S_t) + h^2 \text{Var}(\Delta F_t) - 2h \text{Cov}(\Delta S_t, \Delta F_t)$. Differentiating this expression with respect to h and setting it to zero yields $h^* = \frac{\text{Cov}(\Delta S_t, \Delta F_t)}{\text{Var}(\Delta F_t)}$. Because the covariance between ΔS_t and ΔF_t can be written in terms of the correlation ρ_{SF} and the product of the respective standard deviations σ_S and σ_F , the hedge ratio that minimizes variance becomes $h^* = \rho_{SF} \frac{\sigma_S}{\sigma_F}$.

To connect this result with the linear regression approach, consider a simple ordinary least squares (OLS) regression of ΔS_t on ΔF_t . Here, we often write

$$\Delta S_t = \alpha + \beta \Delta F_t + \varepsilon_t$$

where α is an intercept, β is a slope coefficient, and ε_t is the error term. It is a standard OLS result that the optimal estimated β is given by the ratio of the sample covariance of ΔS_t and ΔF_t to the sample variance of ΔF_t . Specifically, $\beta = \frac{\text{Cov}(\Delta S_t, \Delta F_t)}{\text{Var}(\Delta F_t)}$ by solving the following:

$$\min_{\beta, \alpha} \sum_{t=1}^T (\Delta S_t - \beta \Delta F_t - \alpha)^2$$

In the idealized population limit, this converges to the true ratio of covariance to variance, namely, $\frac{\text{Cov}(\Delta S_t, \Delta F_t)}{\text{Var}(\Delta F_t)}$. That quantity is precisely the hedge ratio derived by minimizing the variance of the hedged portfolio.

The equivalence arises because the slope coefficient in a regression of one random variable on another is the quantity that best explains, in a least squares sense, how changes in the independent variable co-move with changes in the dependent variable. In a hedging context, spot price changes ΔS_t are viewed as the outcome to be “covered,” and futures price changes ΔF_t are the tool used to hedge. The slope β telling us how ΔS_t scales with ΔF_t is the same parameter that appears in the variance-minimizing portfolio, since the variance-minimization problem also requires knowledge of how ΔS_t and ΔF_t move together. Even if the regression contains a nonzero intercept α , that term does not affect the slope and thus does not alter the hedge ratio; the intercept reflects an average drift or level effect that does not change the ratio needed to offset price fluctuations. Hence, whether one directly solves for h^* via differentiation of the hedged portfolio’s variance or estimates β via an OLS regression of ΔS_t on ΔF_t , both frameworks produce the same formula for the optimal hedge ratio in a setting where returns (or price changes) follow a joint normal distribution.

Now, we can come to the dynamic hedging strategy using the futures contract for each period. The process consists of several key steps. First, we simulate price paths for both the asset and futures prices using the GBM model. At each time step, we estimate the optimal hedge ratios through linear regression, enabling the strategy to dynamically adjust hedge positions based on these ratios. Throughout this process,

we calculate the profit and loss for both the asset and the futures positions. Finally, we evaluate the effectiveness of the hedging strategy by comparing the P&L distributions and risk metrics of the hedged and unhedged positions. This comparison provides insights into the performance and risk reduction achieved through the hedging approach.

As shown in Listing 5-1, we begin by importing the necessary libraries and setting simulation parameters, including the time horizon, time steps, and number of simulations.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from sklearn.linear_model import LinearRegression
6
7 # Simulation parameters
8 np.random.seed(42) # For reproducibility
9 T = 0.5             # Time to maturity in years (6 months)
10 dt = 1/252          # Time step (daily)
11 N_steps = int(T / dt) # Number of time steps
12 N_simulations = 5000 # Number of simulated price paths
13
14 # Asset price parameters
15 S0 = 50             # Initial spot price of the asset
16 mu_S = 0.05         # Expected annual return of the asset
17 sigma_S = 0.25      # Annual volatility of the asset
18
19 # Futures price parameters
20 F0 = 50             # Initial futures price
21 mu_F = 0.05         # Expected annual return of the futures
22 sigma_F = 0.25      # Annual volatility of the futures
23
24 # Correlation between asset and futures price changes
25 rho_SF = 0.9
26
27 # Quantity parameters
28 Q = 5000           # Quantity of the asset
29 q = 50             # Contract size of the futures
30 transaction_cost = 0.001 # Transaction cost per unit
```

Listing 5-1 Setting up parameters for simulation

Next, we simulate correlated asset and futures price paths using GBM.

```
1 # Time array
2 time_grid = np.linspace(0, T, N_steps + 1)
3
4 # Preallocate arrays
5 S_paths = np.zeros((N_simulations, N_steps + 1))
6 F_paths = np.zeros((N_simulations, N_steps + 1))
7
8 # Set initial prices
9 S_paths[:, 0] = S0
10 F_paths[:, 0] = F0
11
```



```

12 # Cholesky decomposition for correlated random variables
13 cov_matrix = np.array([[1, rho_SF], [rho_SF, 1]])
14 chol_matrix = np.linalg.cholesky(cov_matrix)
15
16 for i in range(N_simulations):
17     for t in range(1, N_steps + 1):
18         # Generate standard normal random variables
19         z = np.random.normal(size=2)
20         correlated_z = chol_matrix @ z
21
22         # Simulate asset price
23         S_paths[i, t] = S_paths[i, t - 1] * np.exp(
24             (mu_S - 0.5 * sigma_S**2) * dt + sigma_S *
25             correlated_z[0] * np.sqrt(dt)
26         )
27
28         # Simulate futures price
29         F_paths[i, t] = F_paths[i, t - 1] * np.exp(
30             (mu_F - 0.5 * sigma_F**2) * dt + sigma_F *
31             correlated_z[1] * np.sqrt(dt)
32         )

```

Listing 5-2 Simulating GBM processes

Here, we used the Cholesky decomposition to generate correlated random variables, ensuring that the simulated asset and futures prices reflect the desired correlation. At each time step, we can estimate the optimal hedge ratio using linear regression.

```

1 hedge_ratios = []
2
3 for t in range(N_steps):
4     # Calculate price changes
5     delta_S = S_paths[:, t + 1] - S_paths[:, t]
6     delta_F = F_paths[:, t + 1] - F_paths[:, t]
7
8     # Reshape delta_F for regression
9     delta_F = delta_F.reshape(-1, 1)
10
11     # Perform linear regression
12     model = LinearRegression().fit(delta_F, delta_S)
13     h_star = model.coef_[0]
14
15     hedge_ratios.append(h_star)

```

Listing 5-3 Estimating optimal hedge ratio via linear regression

We then roll out the dynamic hedging process using futures contracts by adjusting the futures position on the basis of the estimated optimal hedge ratios.

```

1 # Initialize arrays
2 position_futures = np.zeros((N_simulations, N_steps + 1))
3 cash_flows = np.zeros((N_simulations, N_steps + 1))
4
5 for t in range(N_steps):
6     h_star = hedge_ratios[t]

```

```

9     N_star = -h_star * (Q / q)
10
11     if t == 0:
12         # Initial position
13         position_futures[:, t] = N_star
14     else:
15         # Adjust position
16         delta_N = N_star - position_futures[:, t - 1]
17         position_futures[:, t] = N_star
18
19         # Transaction costs
20         cash_flows[:, t] -= np.abs(delta_N * F_paths[:, t] * q *
21                                     transaction_cost)

```

Listing 5-4 Rolling out dynamic hedge using futures contract

In this code, we calculate the number of futures contracts needed at each time step and account for transaction costs incurred due to adjustments. We also compute the P&L for both the hedged and unhedged positions.

```

1 # Final futures position remains the same as the last adjustment
2 position_futures[:, -1] = position_futures[:, -2]
3
4 # P&L from spot position
5 pnl_spot = Q * (S_paths[:, -1] - S0)
6
7 # P&L from futures position
8 pnl_futures = position_futures[:, -1] * q * (F_paths[:, -1] -
9         F_paths[:, 0])
10
11 # Total transaction costs
12 total_transaction_costs = np.sum(cash_flows, axis=1)
13
14 # Total P&L
15 pnl_hedged = pnl_spot + pnl_futures + total_transaction_costs
16 pnl_unhedged = pnl_spot

```

Listing 5-5 Comparing PnL of hedged and unhedged positions

We then calculate common downside risk metrics such as Value at Risk (VaR) and Expected Shortfall (ES).

```

1 def calculate_risk_metrics(pnl, confidence_level=0.95):
2     pnl_sorted = np.sort(pnl)
3     index = int((1 - confidence_level) * len(pnl_sorted))
4     VaR = -pnl_sorted[index]
5     ES = -np.mean(pnl_sorted[:index])
6     return VaR, ES
7
8 # Hedged position risk metrics
9 VaR_hedged, ES_hedged = calculate_risk_metrics(pnl_hedged)
10
11 # Unhedged position risk metrics
12 VaR_unhedged, ES_unhedged = calculate_risk_metrics(pnl_unhedged)

```

Listing 5-6 Calculating downside risk measures

Now we can visualize the P&L distributions and the hedge ratios over time.

```
1 # P&L Distribution Plot
2 plt.figure(figsize=(12, 6))
3 sns.kdeplot(pnl_unhedged, label='Unhedged Position', shade=True)
4 sns.kdeplot(pnl_hedged, label='Hedged Position', shade=True)
5 plt.title('P&L Distribution of Hedged vs. Unhedged Positions')
6 plt.xlabel('Profit and Loss ($)')
7 plt.ylabel('Density')
8 plt.legend()
9 plt.show()
10
11 # Hedge Ratios Over Time
12 plt.figure(figsize=(10, 4))
13 plt.plot(np.arange(N_steps) * dt, hedge_ratios)
14 plt.title('Optimal Hedge Ratio Over Time')
15 plt.xlabel('Time (Years)')
16 plt.ylabel('Hedge Ratio')
17 plt.show()
```

Listing 5-7 Plotting P&L and hedge ratios

The P&L distribution plot (left panel of Figure 5-1) shows that the hedged position has less variability compared to the unhedged position, indicating a reduced risk. The hedge ratio plot (right panel of Figure 5-1) illustrates how the hedge ratio fluctuates over time, reflecting adjustments made in response to changing market conditions.

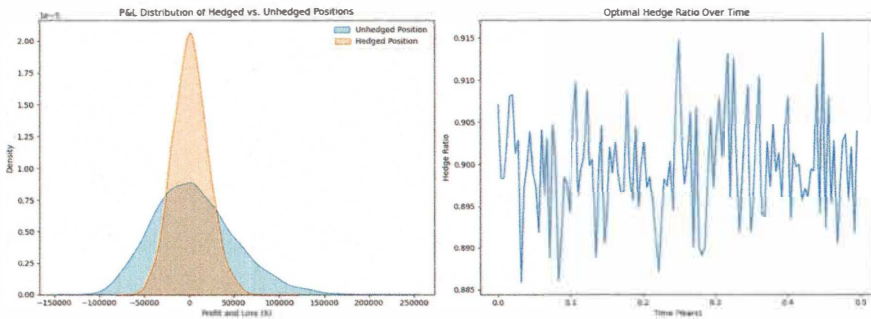


Figure 5-1 Illustrating hedging effectiveness. The left panel shows the P&L distribution of hedged and unhedged positions. The right panel shows the estimated optimal hedge ratio over time

Finally, we print the risk metrics as shown in Listing 5-8.

```
1 print(f"Hedged Position VaR at 95% confidence: ${VaR_hedged:,.2f}"
2      ")
3 print(f"Hedged Position Expected Shortfall at 95% confidence: ${
4      ES_hedged:,.2f}\n")
5 print(f"Unhedged Position VaR at 95% confidence: ${VaR_unhedged
6      :,.2f}")
7 print(f"Unhedged Position Expected Shortfall at 95% confidence: $
8      {ES_unhedged:,.2f}")
9
10 # Output
11 Hedged Position VaR at 95% confidence: $31,761.44
12 Hedged Position Expected Shortfall at 95% confidence: $39,471.00
13
14 Unhedged Position VaR at 95% confidence: $63,486.41
15 Unhedged Position Expected Shortfall at 95% confidence: $75
16     ,266.29
```

Listing 5-8 Printing risk metrics

The results indicate that a hedged position has considerably lower VaR and ES compared to an unhedged position, which means a reduction in risk. By adopting a dynamic hedging strategy that adjusts the hedge ratio over time based on current market data, we can more effectively manage risk in changing market conditions.

5.2 Hedging with Option Contracts

Option contracts are a common instrument for hedging and offer greater flexibility and adaptability than futures contracts. As mentioned earlier, an option gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price within a specific time frame (typically on or before the expiration date). An option contract is a type of derivative, which means that its value is derived on the basis of an underlying asset such as stocks, commodities, or currencies. In this section, we explore how options can be used for hedging, describe key hedging strategies, and offer detailed frameworks and examples to illustrate their practical usage. We will focus mainly on European options.

There are two main types of options: call options and put options. A call option allows the holder to buy the underlying asset at a predetermined price known as the strike price (denoted K) on the expiration date. In contrast, a put option grants the holder the right to sell the underlying asset at the strike price K on the expiration date. The cost of purchasing an option is referred to as the premium paid to the other party (ignoring middlemen such as exchanges or brokers). Options can be traded on exchanges, where contracts are standardized, or in over-the-counter markets, where contracts can be customized to meet specific needs.

Options present particularly versatile hedging opportunities by virtue of their asymmetric payoff profiles, which allow potential gains and losses to be managed differently compared to linear instruments like futures. Two widely used option-based hedging strategies are the protective put and the covered call. The protective

put involves buying a put option on an asset already held, thereby placing a “floor” on potential losses if the asset price falls below the strike. Meanwhile, the covered call involves selling (writing) a call option on an asset that one owns, generating immediate premium income but capping upside gains if the asset’s price rises substantially. Both of these strategies illustrate how options can be used to fine-tune a portfolio’s risk-return profile, balancing downside risk mitigation against foregone opportunities for higher profits.

These hedging techniques harness the mathematically distinctive payoff structures of options to manage risk in a precise and adaptable manner. By carefully selecting the strike price and accounting for the option premium, we can construct tailored hedges that align with specific risk preferences, market outlooks, and time horizons. Consequently, the resulting strategies offer both reliable downside protection and the flexibility to accommodate different market conditions, making them a powerful tool for controlling exposure to adverse price movements.

Let us look at these two common hedging strategies in detail, starting with the protective put strategy.

5.2.1 Protective Put Strategy

The protective put strategy serves as a mechanism to protect against potential declines in the value of an asset while preserving the opportunity to gain should the value of the asset appreciate. Conceptually, it mirrors the principle behind futures-based hedges, but with the added flexibility due to the asymmetric payoff of the option. Specifically, say that we have Q units of an asset priced at S_0 and wish to shield the investment from downside risk at a future time T . We can purchase N put option contracts, each covering q units, at a chosen strike price K . This approach effectively places a floor limit on how much we can lose if the asset price falls below K , while allowing for upside potential if market prices rise. The trade-off is the premium paid for the put options, which slightly reduces any potential gain. Mathematically, the total initial outlay, reflecting both the spot holding and the premium expenditure, is given by

$$\text{Initial Outlay} = Q \times S_0 + N \times q \times \text{Premium}_{\text{put}},$$

where the terms ensure that the hedging cost is applicable and the downside protection is in place. Here, the premium is calculated per share of the underlying asset. At the expiration date of the option T , the value of our protective put position depends on the price of the asset at that time S_T . The payoff from holding the asset is

$$\text{Payoff}_{\text{asset}} = Q \times (S_T - S_0)$$

which represents the gain or loss due to ownership of the asset over the period.

On the other hand, the payoff per share from each put option is

$$\text{Payoff}_{\text{put}} = \max(K - S_T, 0)$$

So, the total payoff from all $N \times q$ put options is

$$\text{Total Payoff}_{\text{put}} = N \times q \times \max(K - S_T, 0)$$

Therefore, the total profit or loss $\Pi_{\text{protective put}}$ of our portfolio at maturity (time T) is

$$\Pi_{\text{protective put}} = \text{Payoff}_{\text{asset}} + \text{Total Payoff}_{\text{put}} - N \times q \times \text{Premium}_{\text{put}}$$

This formula calculates the net profit or loss from the hedged position, considering the change in the asset's value, the payoff from the put options, and the premium paid.

Note that an equivalent way to express the protective put payoff is to compare the final value of the combined position to the initial outlay. Specifically, at maturity, the final value consists of two parts: the value of the underlying asset, $Q S_T$, plus the value of the put options, $N q \max(K - S_T, 0)$. The total initial outlay is simply the cost of acquiring the asset, $Q S_0$, plus the premium paid for the puts, $N q \text{Premium}_{\text{put}}$. Thus, we can write

$$\Pi_{\text{protective put}} = \underbrace{(Q S_T + N q \max(K - S_T, 0))}_{\text{Final Value}} - \underbrace{(Q S_0 + N q \text{Premium}_{\text{put}})}_{\text{Initial Outlay}},$$

which is mathematically equivalent to the previous expression as the sum of the change in the price of the asset plus the return of the put and minus the premium of the put.

The protective put strategy effectively sets a minimum price level (the strike price K minus the premium paid per share) below which you will not lose more money, while still allowing for unlimited gains if the asset's price rises. In other words, the protective put strategy offers a form of "insurance" for the asset, ensuring that its value cannot drop below a certain floor while preserving the upside potential. By holding the underlying asset and simultaneously purchasing put options with strike price K , the investor gains the contractual right (but not the obligation) to sell the asset at K . In practice, this creates a lower limit on losses: if the asset's market price falls well below K , put options can be exercised (or sold at an intrinsic value) to offset further declines, effectively limiting downside. The main cost of this downside protection is the option premium, which reduces the net payoff by a fixed amount per share. Consequently, if the asset's price is above K at expiration, the put options can expire worthless, allowing the investor to capture all the upside gains of the asset minus the initial premium paid.

As shown in Figure 5-2, a put option produces a skewed payoff line that flattens below K (reflecting limited loss) and rises in tandem with the asset price above K (preserving the profit potential). Hence, the strategy secures a minimum effective selling price around $K - \text{Premium}$ per share, while still offering open-ended gains if the market moves favorably. The graph illustrates the payoff curves for three

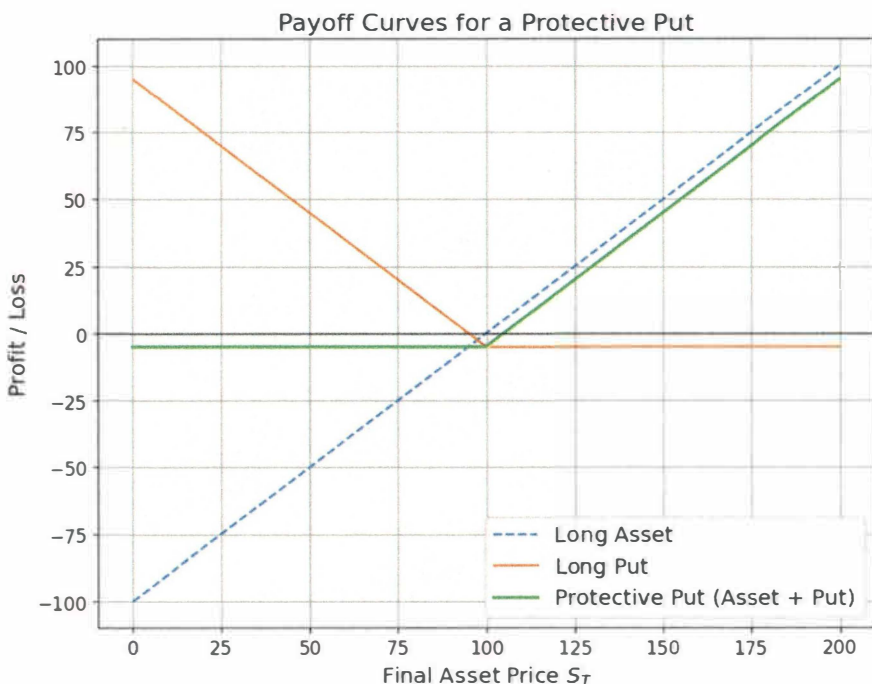


Figure 5-2 P&L of a protective put option

positions: (1) owning the underlying asset alone, (2) buying a put option, and (3) combining both into a protective put. The asset payoff we own has a slope of +1 (i.e., the profit or loss increases one-for-one with the underlying's price S_T), crossing the horizontal axis at $S_T = S_0$. The put payoff remains at $-\text{Premium}$ as long as $S_T > K$ (since the option expires worthless in that region), then slopes upward as $K - S_T - \text{Premium}$ when $S_T < K$. When these two positions are combined, the resulting protective put payoff matches the asset payoff above the strike K but flattens out below K , effectively placing a floor on losses. Specifically, for $S_T < K$, the intrinsic value of the put compensates for any further decline in the asset's price, limiting downside exposure to approximately $(K - S_0 - \text{Premium})$. Above K , the position behaves much like a simple long stock, retaining unlimited upside minus the fixed cost of the put premium. This asymmetry, limited loss, and open-ended gains make a protective put an attractive hedging strategy for investors who wish to preserve significant upside potential while insuring against catastrophic losses.

Listing 5-9 is used to generate the plot.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Adjust global font size for all text elements
5 plt.rcParams.update({
6     'font.size': 14,          # Main text font size
```

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Listing 5-9 Generating payoff curves for the protective put strategy

For example, suppose we own 500 shares ($Q = 500$) of a stock currently priced at $S_0 = \$80$ per share. We are concerned about a potential decline in the stock price over the next few months. To hedge this risk, we can purchase put options with a strike price of $K = \$80$. Assume the premium per share is $\text{Premium}_{\text{put}} = \3 . Since each option contract typically covers 100 shares ($q = 100$), we need

$$N = \frac{Q}{q} = \frac{500}{100} = 5 \text{ contracts}$$

The total premium paid is

$$\text{Total Premium}_{\text{put}} = N \times q \times \text{Premium}_{\text{put}} = 5 \times 100 \times \$3 = \$1,500$$

The total initial investment outlay is

$$\text{Initial Outlay} = 500 \times \$80 + 5 \times 100 \times \$3 = \$40,000 + \$1,500 = \$41,500$$

If the stock price falls to $S_T = \$65$, the payoff from the asset is

$$\text{Payoff}_{\text{asset}} = 500 \times (\$65 - \$80) = 500 \times (-\$15) = -\$7,500$$

which represents a loss due to the decrease in the stock price.

Next, the payoff from the put options is:

$$\text{Total Payoff}_{\text{put}} = 5 \times 100 \times (\$80 - \$65) = 500 \times \$15 = \$7,500$$

These put options are “in the money” and thus provide a gain that offsets the loss from the asset.

The total profit or loss becomes

$$\Pi_{\text{protective put}} = -\$7,500 + \$7,500 - \$1,500 = -\$1,500$$

which shows that the net loss is equal to the premium paid for the put options. In other words, the loss from the asset’s price decline is fully offset by the gain from the put options, except for the premium paid.

On the other hand, if the stock price rises to $S_T = \$90$, the payoff from the asset becomes

$$\text{Payoff}_{\text{asset}} = 500 \times (\$90 - \$80) = 500 \times \$10 = \$5,000$$

Table 5-1 Payoff and profit/loss table

Stock price at maturity (S_T)	Payoff from asset	Payoff from put options	Premium paid	Total P&L
\$65	− \$7,500	\$7,500	− \$1,500	− \$1,500
\$90	\$5,000	\$0	− \$1,500	\$3,500

which is the gain from the increase in the stock price. As for the payoff of the put options, since $S_T > K$, the put options expire worthless:

$$\text{Total Payoff}_{\text{put}} = 0$$

Thus, the total profit or loss is

$$\Pi_{\text{protective put}} = \$5,000 + \$0 - \$1,500 = \$3,500$$

The net profit is the gain from the asset minus the premium paid for the put options. Thus, we benefit from the asset's price increase, but incur the cost of the premium.

Table 5-1 summarizes the P&L of each component and the overall portfolio of the protective put strategy in each scenario. This strategy essentially limits our downside risk to the premium paid while allowing for upside potential if the asset price increases.

The next section looks at implementing this strategy and observing its effect via simulations.

5.2.2 Implementing the Protective Put Strategy

Now that we understand the protective put strategy, let us implement it using Python. We will simulate asset price movements, calculate option premiums, and evaluate payoffs to see how effective the hedge is in different scenarios. We will again use GBM to simulate asset price movements and the Black-Scholes-Merton (BSM) model to price the options. As introduced earlier, GBM can be used to capture both the predictable trend (drift) and random fluctuations (volatility) when simulating how the asset price changes over time. Recall that we discretize the GBM using the following equation:

$$S_{t+\Delta t} = S_t \times \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) \Delta t + \sigma \epsilon_t \sqrt{\Delta t} \right)$$

where Δt is the time step size, and ϵ_t is a standard normal random variable ($\epsilon_t \sim N(0, 1)$).

To price European put options, we can use the Black-Scholes-Merton model:

$$P = Ke^{-r(T-t)}N(-d_2) - S_tN(-d_1)$$

where

$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

In these equations, $N(\cdot)$ is the cumulative distribution function of the standard normal distribution, S_t is the asset price at time t , K is the strike price, r is the risk-free interest rate, and T is the time to maturity. Recall that at maturity T , the payoff of the put option is

$$\text{Payoff}_{\text{put}} = \max(K - S_T, 0)$$

We can now simulate the protective put strategy by modeling asset price paths using GBM, calculating option premiums with the BSM model, and evaluating the payoffs from both the asset and the put options.

First, we import the necessary packages and define the parameters for our simulation as shown in Listing 5-10.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from scipy.stats import norm
5
6 # Simulation parameters
7 np.random.seed(42)           # For reproducibility
8 S0 = 100                     # Initial asset price ($100)
9 K = 100                       # Strike price of the put option ($100)
10 T = 0.5                      # Time to maturity in years (6 months)
11 r = 0.02                     # Risk-free interest rate (2%)
12 sigma = 0.25                 # Volatility of the asset (25%)
13 mu = 0.10                    # Expected return of the asset (10%)
14 N_simulations = 5000         # Number of simulation paths
15 N_steps = 126                # Number of time steps (approx. 6
    months of trading days)
16 dt = T / N_steps             # Time step size
17 Q = 2000                     # Quantity of the asset held
18 q = 100                      # Quantity per option contract
19 N_options = Q // q           # Number of option contracts
```

Listing 5-10 Defining simulation parameters

Next, we calculate the premium of the put option at time $t = 0$.

```

1 def black_scholes_put(S, K, T, r, sigma):
2     d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np
3         .sqrt(T))
4     d2 = d1 - sigma * np.sqrt(T)
5     put_price = K * np.exp(-r * T) * norm.cdf(-d2) - S * norm.cdf
6         (-d1)
7     return put_price
8
9 # Calculate the premium per option
10 premium_per_option = black_scholes_put(S0, K, T, r, sigma)
11 print(f"Premium per Put Option: ${premium_per_option:.2f}")
12
13 # Total premium paid
14 total_premium = N_options * q * premium_per_option
15 print(f"Total Premium Paid: ${total_premium:.2f}")
16
17 # Output
18 Premium per Put Option: $6.52
19 Total Premium Paid: $13043.66

```

Listing 5-11 Calculating option premium

Now we can simulate multiple asset price paths over the time horizon T .

```

1 # Time array
2 time_grid = np.linspace(0, T, N_steps + 1)
3
4 # Preallocate array for asset prices
5 S_paths = np.zeros((N_simulations, N_steps + 1))
6 S_paths[:, 0] = S0
7
8 # Simulate price paths
9 for t in range(1, N_steps + 1):
10     Z = np.random.standard_normal(N_simulations)
11     S_paths[:, t] = S_paths[:, t - 1] * np.exp((mu - 0.5 * sigma
12         **2) * dt + sigma * Z * np.sqrt(dt))

```

Listing 5-12 Simulating asset price paths

At maturity, we can calculate the payoffs from both the asset and the put options.

```

1 # Asset price at maturity
2 S_T = S_paths[:, -1]
3
4 # Payoff from the asset
5 payoff_asset = Q * (S_T - S0)
6
7 # Payoff from the put options
8 payoff_put = N_options * q * np.maximum(K - S_T, 0)
9
10 # Total profit or loss
11 total_pnl = payoff_asset + payoff_put - total_premium

```

Listing 5-13 Calculating payoffs

Now we can analyze the hedging effectiveness by comparing the protective put strategy to holding the asset without hedging as shown in Listing 5-14.

```

1 # Profit or loss without hedging
2 pnl_unhedged = payoff_asset
3
4 # Create a DataFrame for comparison
5 results = pd.DataFrame({
6     'Asset_Price_T': S_T,
7     'Payoff_Asset': payoff_asset,
8     'Payoff_Put': payoff_put,
9     'Total_PnL_Hedged': total_pnl,
10    'PnL_Unhedged': pnl_unhedged
11 })
12
13 # Calculate descriptive statistics
14 stats = results[['Total_PnL_Hedged', 'PnL_Unhedged']].describe()
15 print(stats)
16
17 # Output
18 count    5000.000000    5000.000000
19 mean      6833.195431    9698.512913
20 std      26862.596609    37183.472059
21 min     -13043.659498   -91149.304063
22 25%     -13043.659498   -17461.066383
23 50%     -6159.277399     6884.382098
24 75%     19882.797212    32926.456709
25 max      178995.442071   192039.101569

```

Listing 5-14 Comparing strategies

We can also plot the distribution of the profits and losses for both hedged and unhedged positions:

```

1 # Plot P&L distributions
2 plt.figure(figsize=(12, 6))
3 plt.hist(total_pnl, bins=100, alpha=0.6, label='Hedged Position')
4 plt.hist(pnl_unhedged, bins=100, alpha=0.6, label='Unhedged
   Position')
5 plt.title('Distribution of Profit and Loss', fontsize=18)
6 plt.xlabel('Profit and Loss ($)', fontsize=16)
7 plt.ylabel('Frequency', fontsize=16)
8 plt.legend(fontsize=16)
9 plt.show()

```

The plot, as shown in Figure 5-3, shows that the hedged position reduces downside risk, with losses capped at the premium paid. The unhedged position has a wider distribution, indicating higher potential losses and gains.

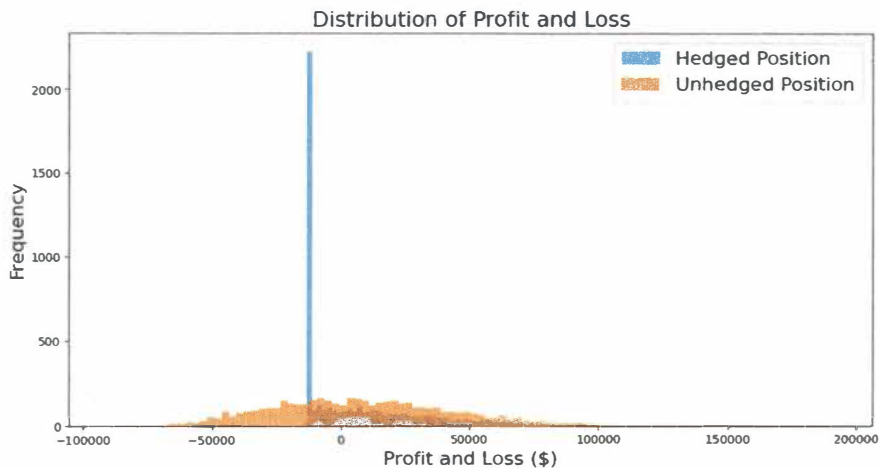


Figure 5-3 Distribution of profit and loss for hedged vs. unhedged positions

Finally, we examine specific scenarios to see how the protective put strategy performs under different market conditions:

```

1 # Scenario where asset price falls to $70
2 scenario_loss = results[results['Asset_Price_T'] <= 70].iloc[0]
3 print("\nScenario: Asset price falls to $70")
4 print(f"Asset Price at Maturity: ${scenario_loss['Asset_Price_T']:.2f}")
5 print(f"Payoff from Asset: ${scenario_loss['Payoff_Asset']:.2f}")
6 print(f"Payoff from Put Options: ${scenario_loss['Payoff_Put']:.2f}")
7 print(f"Total P&L (Hedged): ${scenario_loss['Total_PnL_Hedged']:.2f}")
8 print(f"P&L (Unhedged): ${scenario_loss['PnL_Unhedged']:.2f}")
9
10 # Scenario where asset price rises to $130
11 scenario_gain = results[results['Asset_Price_T'] >= 130].iloc[0]
12 print("\nScenario: Asset price rises to $130")
13 print(f"Asset Price at Maturity: ${scenario_gain['Asset_Price_T']:.2f}")
14 print(f"Payoff from Asset: ${scenario_gain['Payoff_Asset']:.2f}")
15 print(f"Payoff from Put Options: ${scenario_gain['Payoff_Put']:.2f}")
16 print(f"Total P&L (Hedged): ${scenario_gain['Total_PnL_Hedged']:.2f}")
17 print(f"P&L (Unhedged): ${scenario_gain['PnL_Unhedged']:.2f}")
18
19 # Output
20 Scenario: Asset price falls to $70
21 Asset Price at Maturity: $61.64
22 Payoff from Asset: $-76714.94
23 Payoff from Put Options: $76714.94

```

```

24 Total P&L (Hedged): $-13043.66
25 P&L (Unhedged): $-76714.94
26
27 Scenario: Asset price rises to $130
28 Asset Price at Maturity: $140.75
29 Payoff from Asset: $81504.28
30 Payoff from Put Options: $0.00
31 Total P&L (Hedged): $68460.62
32 P&L (Unhedged): $81504.28

```

In the scenario where the asset price falls to \$70, the loss from the asset is fully offset by the gain from the put options, except for the premium paid, resulting in a net loss equal to the premium. In the scenario where the asset price rises to \$130, the gain from the asset is reduced by the premium paid for the put options, but we still achieve a significant profit. By examining these scenarios, we see how the protective put strategy limits losses while still allowing for potential gains, offering a practical method to manage risk.

Next, we turn to the covered call strategy.

5.2.3 Covered Call Strategy

A covered call strategy involves owning the underlying asset and simultaneously writing (selling) call options on the same asset. Also called a yield-enhancement strategy, this approach can generate additional income by receiving the option premium, but limit the upside if the asset price rises well above the strike price. The received premium can act as a buffer against minor price declines in the underlying asset, although protection is limited and risk mitigation is only partial. If the underlying asset drops significantly, the writer still assumes most of the downside risk beyond the premium.

Specifically, suppose that we have Q units of an asset priced at S_0 . To implement a covered call, we write (sell) N call option contracts, each covering q units (each standard contract typically covers $q = 100$ shares), at a chosen strike price K . Because we receive a premium for writing these calls, we earn extra cash flow up front, which can partially offset minor price declines in the underlying asset. However, should the underlying price increase substantially above K , our additional gains on the asset are capped by the short call position, making this a trade-off between enhanced current yield and limited upside potential.

Since we receive the call premium rather than pay it, the net initial outlay is

$$\text{Initial Outlay} = Q S_0 - (N \times q \times \text{Premium}_{\text{call}}),$$

which reflects the fact that the call premium provides an immediate cash inflow that partially offsets the asset's purchase cost. Often, one chooses $N = \frac{Q}{q}$ so that every share held in the spot market is "covered" by exactly one corresponding short call share.

Now, let S_T be the asset's price at option expiration T . The covered call strategy's P&L (the terminal payoff) then depends on two components: asset in a long position and call option in a short position. For the long asset payoff, if we purchased Q shares at S_0 , the change in the asset's value over the period is

$$\text{Payoff}_{\text{asset}} = Q (S_T - S_0).$$

For the short call payoff, writing the call grants the buyer the right to purchase the asset at price K . If $S_T \leq K$, the option expires worthless, and our payoff from the short call position is simply the premium received: $Nq \text{ Premium}_{\text{call}}$. However, if $S_T > K$, the call is exercised, and we must deliver the asset at K . The intrinsic loss on the short call is $\max(S_T - K, 0)$ per share, multiplied by Nq total shares. Factoring in the premium, the net payoff from the short call can be expressed as

$$\text{Payoff}_{\text{short call}} = Nq \text{ Premium}_{\text{call}} - Nq \max(S_T - K, 0).$$

Combining these two payoffs gives the covered call portfolio's P&L at maturity:

$$\Pi_{\text{covered call}} = \underbrace{Q (S_T - S_0)}_{\text{Asset}} + \underbrace{[Nq \text{ Premium}_{\text{call}} - Nq \max(S_T - K, 0)]}_{\text{Short Call}}.$$

Again, an equivalent way to write this is to compare the final value of the portfolio to the initial outlay:

$$\Pi_{\text{covered call}} = \underbrace{(Q S_T - Nq \max(S_T - K, 0))}_{\text{Final Value}} - \underbrace{(Q S_0 - Nq \text{ Premium}_{\text{call}})}_{\text{Initial Outlay}},$$

which clarifies that any gains above K on the asset are effectively “given up” to the call buyer, while the premium is ours to keep regardless of the outcome.

In essence, the short call “covers” part of our cost by adding immediate premium income, thus providing a small buffer against price declines. On the other hand, once S_T rises above K , we cease to benefit from additional upside, as we must sell the asset at K . However, below K , we bear the downside risk almost in full (minus the premium). As a result, a covered call strategy is moderately bullish, in the sense that

- It performs best in scenarios where the underlying price stays below or moderately above K .
- If the asset surges well beyond K , our gains become capped relative to holding the asset alone.
- If the asset plummets, we still incur losses similar to an unhedged position, offset slightly by the option premium.

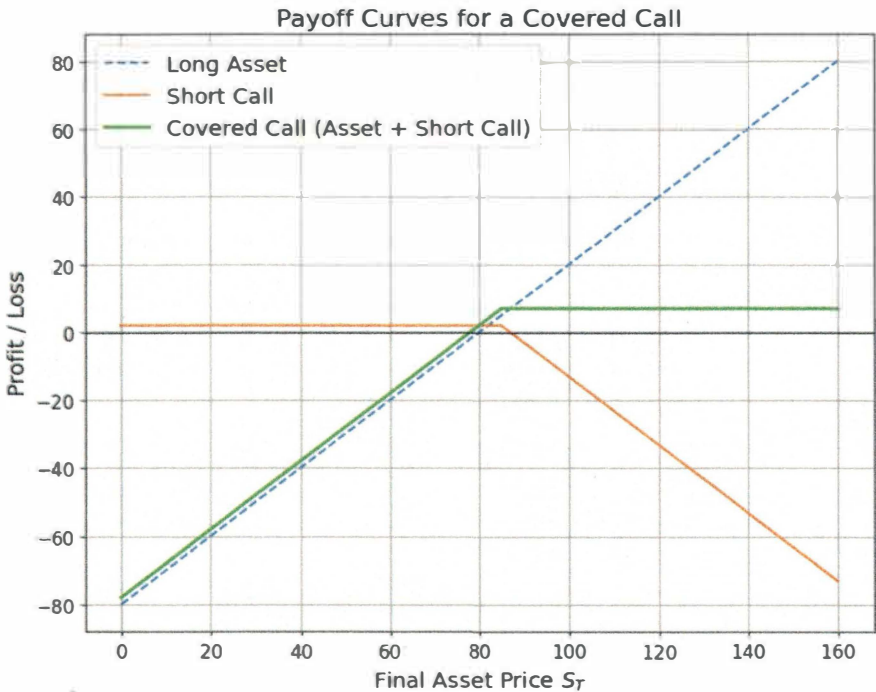


Figure 5-4 Payoff diagram for a covered call strategy compared to its components

Figure 5-4 presents the payoff diagram that consists of a long asset alone, a short call alone, and the combined covered call. For the short call position, the payoff is simply $\text{Premium} - \max(S_T - K, 0)$, which remains at the premium received when $S_T \leq K$ and declines linearly for $S_T > K$. The covered call adds this short-call payoff to the long-asset payoff, resulting in a payoff curve that is identical to the long-asset payoff up to $S_T = K$, then flattens out beyond K . That flat portion indicates the foregone upside. Meanwhile, the initial premium from writing the call acts as a cushion against mild price dips.

Listing 5-15 shows how to generate this graph.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Adjust global font size for all text elements
5 plt.rcParams.update({
6     'font.size': 14,          # Main text font size
7     'axes.titlesize': 16,     # Axes title font size
8     'axes.labelsize': 14,     # Axes label font size
9     'legend.fontsize': 14,    # Legend font size
10    'xtick.labelsize': 12,     # X tick label size
11    'ytick.labelsize': 12      # Y tick label size
12 })

```

```

13
14 # Parameters
15 S0 = 100.0          # Initial price of the asset
16 K = 100.0          # Put option strike
17 put_premium = 5     # Cost of the put option (per share)
18
19 # Generate a range of possible final asset prices (S_T)
20 S_T = np.linspace(0, 200, 201)
21
22 # Payoff 1: Long asset (bought at S0).
23 #     P&L = (S_T - S0)
24 payoff_asset = S_T - S0
25
26 # Payoff 2: Long put option
27 #     Gross payoff = max(K - S_T, 0)
28 #     Net payoff after premium = max(K - S_T, 0) - put_premium
29 payoff_put = np.maximum(K - S_T, 0) - put_premium
30
31 # Combined payoff: Protective put = Long asset + Long put
32 payoff_protective = payoff_asset + payoff_put
33
34 # Plot the results
35 plt.figure(figsize=(9, 7))
36
37 # Plot each component payoff
38 plt.plot(S_T, payoff_asset, label='Long Asset', linestyle='--')
39 plt.plot(S_T, payoff_put, label='Long Put')
40 plt.plot(S_T, payoff_protective, label='Protective Put (Asset +
    Put)', linewidth=2)
41
42 # Reference line at zero P&L
43 plt.axhline(y=0, color='black', linewidth=1)
44
45 # Labeling
46 plt.xlabel('Final Asset Price $S_T$')
47 plt.ylabel('Profit / Loss')
48 plt.title('Payoff Curves for a Protective Put')
49 plt.legend()
50 plt.grid(True)
51
52 # Display the figure
53 plt.show()

```

Listing 5-15 Generating payoff curves for covered call strategy

Let us look at a specific example. Suppose we hold $Q = 500$ shares of a stock currently priced at $S_0 = \$80$. We sell call options at strike $K = \$85$, each option covering $q = 100$ shares. Assume the call premium per share is $\text{Premium}_{\text{call}} = \2 . The required number of call contracts to cover 500 shares is

$$N = \frac{Q}{q} = \frac{500}{100} = 5 \text{ contracts.}$$

The total premium received from writing these five contracts is

$$\text{Total Premium}_{\text{call}} = N \times q \times \text{Premium}_{\text{call}} = 5 \times 100 \times \$2 = \$1,000.$$

Hence, our net initial outlay is the cost of buying (or holding) the 500 shares minus the premium income:

$$\text{Initial Outlay} = 500 \times \$80 - \$1,000 = \$40,000 - \$1,000 = \$39,000.$$

Now consider two scenarios at option expiration. When $S_T = \$90$ (above the strike), the short calls will be exercised, forcing us to sell the stock at \$85 per share. We still own 500 shares, but our effective sale price is \$85. The resulting payoffs will be

- $\text{Payoff}_{\text{asset}} = 500 \times (85 - 80) = \$2,500.$
- $\text{Payoff}_{\text{short call}} = +\$1,000 - 500 \times (90 - 85) = +\$1,000 - \$2,500 = -\$1,500.$
- $\text{Total P\&L} = \$2,500 + (-\$1,500) = \$1,000.$

Thus, compared to simply holding the shares (which would have gained \$5,000 if sold at \$90), our extra income is capped once S_T exceeds \$85, resulting in only \$1,000 net profit plus the difference in initial outlay.

When $S_T = \$75$ (below the strike), the calls expire worthless ($S_T < K$), so we keep the shares plus the entire \$1,000 premium. The resulting payoffs will be

- $\text{Payoff}_{\text{asset}} = 500 \times (75 - 80) = -\$2,500.$
- $\text{Payoff}_{\text{short call}} = +\$1,000 - 0 = +\$1,000.$
- $\text{Total P\&L} = -\$2,500 + \$1,000 = -\$1,500.$

The short call cushions our loss by \$1,000, but we still incur a net loss if the stock falls significantly.

Table 5-2 summarizes these outcomes. While the call premium provides some downside protection, large price drops still lead to notable losses, and substantial price increases only yield limited gains once S_T exceeds the strike K .

In general, a covered call strategy suits investors with a neutral to moderately bullish outlook who wish to improve the yield through premium income but accept a cap on gains if the asset's price rises beyond strike. Although it offers partial (limited) protection against small price declines, it leaves the investor exposed to large downward moves aside from the modest premium cushion.

Table 5-2 Covered call payoff in two scenarios

S_T	Asset P&L	Short call P&L	Net premium	Total P&L
\$75	-\$2,500	\$0	\$1,000	-\$1,500
\$90	\$2,500	-\$2,500	\$1,000	\$1,000

5.2.4 Implementing the Covered Call Strategy

This section demonstrates how to implement the covered call strategy. As introduced earlier, the covered call strategy involves owning an underlying asset and simultaneously writing call options on it to earn premium income. Although this strategy improves returns when the asset price remains stable or increases moderately, it also caps potential profits if the price rises well above the strike price.

Consider owning $Q = 1,000$ shares of a stock currently priced at $S_0 = \$70$ per share. We write call options with a strike price $K = \$75$. Since each contract covers $q = 100$ shares, the number of contracts is $N = Q/q = 1,000/100 = 10$. We assume a time to maturity of $T = 1$ year, a risk-free interest rate $r = 0.03$, a volatility $\sigma = 0.3$, and an expected return of the asset $\mu = 0.12$. We simulate $N_{\text{simulations}} = 5,000$ price paths over $N_{\text{steps}} = 252$ time steps, resulting in $\Delta t = T/N_{\text{steps}}$.

The Black-Scholes formula for a European call option gives the call price as

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$\text{Call Price} = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where $N(\cdot)$ is the cumulative distribution function of the standard normal distribution.

The codes in Listing 5-16 is the Python code that implements this strategy. We first calculate the call premium using the Black-Scholes model, then simulate asset price paths under a geometric Brownian motion, compute the terminal payoffs, and finally compare the covered call strategy against holding the asset unhedged.

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from scipy.stats import norm
5
6 # Parameters
7 np.random.seed(42)
8 S0 = 70
9 K = 75
10 T = 1.0
11 r = 0.03
12 sigma = 0.3
13 mu = 0.12
14 N_simulations = 5000
15 N_steps = 252
16 dt = T / N_steps
17 Q = 1000
18 q = 100
19 N_options = Q // q
```

```

21 def black_scholes_call(S, K, T, r, sigma):
22     d1 = (np.log(S / K) + (r + 0.5 * sigma**2)*T) / (sigma * np.
23         sqrt(T))
24     d2 = d1 - sigma * np.sqrt(T)
25     call_price = S * norm.cdf(d1) - K * np.exp(-r*T) * norm.cdf(
26         d2)
27     return call_price
28
29 # Compute the call option premium
30 premium_per_option = black_scholes_call(S0, K, T, r, sigma)
31 total_premium = N_options * q * premium_per_option
32 print(f"Premium per Call Option: ${premium_per_option:.2f}")
33 print(f"Total Premium Received: ${total_premium:.2f}")
34
35 # Simulate asset price paths
36 time_grid = np.linspace(0, T, N_steps + 1)
37 S_paths = np.zeros((N_simulations, N_steps + 1))
38 S_paths[:, 0] = S0
39
40 for t in range(1, N_steps + 1):
41     Z = np.random.standard_normal(N_simulations)
42     S_paths[:, t] = S_paths[:, t-1] * np.exp((mu - 0.5*sigma**2)*
43         dt + sigma*Z*np.sqrt(dt))
44
45 # Compute payoffs at maturity
46 S_T = S_paths[:, -1]
47 payoff_asset = Q * (S_T - S0)
48 payoff_call_seller = -N_options * q * np.maximum(S_T - K, 0)
49 total_pnl = payoff_asset + payoff_call_seller + total_premium
50
51 pnl_unhedged = payoff_asset
52 results = pd.DataFrame({
53     'Asset_Price_T': S_T,
54     'Payoff_Asset': payoff_asset,
55     'Payoff_Call_Seller': payoff_call_seller,
56     'Total_PnL_Covered_Call': total_pnl,
57     'PnL_Unhedged': pnl_unhedged
58 })
59
60 stats = results[['Total_PnL_Covered_Call', 'PnL_Unhedged']].
61     describe()
62 print(stats)
63
64 # Plot the P&L distributions
65 plt.figure(figsize=(12,6))
66 plt.hist(total_pnl, bins=100, alpha=0.6, label='Covered Call
67     Strategy')
68 plt.hist(pnl_unhedged, bins=100, alpha=0.6, label='Unhedged
69     Position')
70 plt.title('Distribution of Profit and Loss', fontsize=18)
71 plt.xlabel('Profit and Loss ($)', fontsize=16)
72 plt.ylabel('Frequency', fontsize=16)
73 plt.legend(fontsize=16)

```

```

68 plt.show()
69
70 # Output
71 Premium per Call Option: $7.19
72 Total Premium Received: $7191.96
73
74      Total_PnL_Covered_Call    PnL_Unhedged
75 count                5000.000000        5000.000000
76 mean                 4688.877238        8350.378168
77 std                 10405.567618        23757.218489
78 min                -39000.394164       -46192.358760
79 25%                 -1173.264159        -8365.228754
80 50%                 12191.964596         5187.967171
81 75%                 12191.964596        21884.387136
82 max                 12191.964596       153288.458197

```

Listing 5-16 Implementing the covered call strategy

The result shows that the mean profit for the covered call strategy is lower than the unhedged position, reflecting the capped upside. The standard deviation is lower for the covered call, indicating reduced volatility. The minimum loss is similar for both, but the covered call provides some income to offset losses. This observation is further illustrated in Figure 5-5.

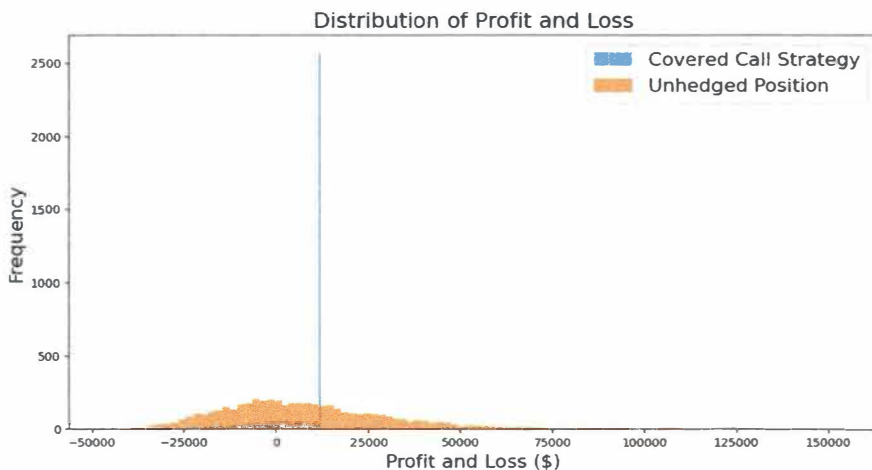


Figure 5-5 Distribution of profit and loss for covered call strategy vs. unhedged position

Finally, let us examine specific scenarios to illustrate the strategy's performance.

```
1 # Scenario where asset price ~ $70
2 scenario_70 = results.iloc[(results['Asset_Price_T'] - 70).abs().
3     argsort()[0]]
4 print("Scenario: Asset price near $70")
5 print(f"Asset Price at Maturity: ${scenario_70['Asset_Price_T']:.2f}")
6 print(f"Payoff from Asset: ${scenario_70['Payoff_Asset']:.2f}")
7 print(f"Payoff from Call Options (Seller): ${scenario_70['Payoff_Call_Seller']:.2f}")
8 print(f"Total P&L (Covered Call): ${scenario_70['Total_PnL_Covered_Call']:.2f}")
9 print(f"P&L (Unhedged): ${scenario_70['PnL_Unhedged']:.2f}")
10
11 # Scenario where asset price ~ $85
12 scenario_85 = results.iloc[(results['Asset_Price_T'] - 85).abs().
13     argsort()[0]]
14 print("\nScenario: Asset price near $85")
15 print(f"Asset Price at Maturity: ${scenario_85['Asset_Price_T']:.2f}")
16 print(f"Payoff from Asset: ${scenario_85['Payoff_Asset']:.2f}")
17 print(f"Payoff from Call Options (Seller): ${scenario_85['Payoff_Call_Seller']:.2f}")
18 print(f"Total P&L (Covered Call): ${scenario_85['Total_PnL_Covered_Call']:.2f}")
19 print(f"P&L (Unhedged): ${scenario_85['PnL_Unhedged']:.2f}")
20
21 # Output
22 Scenario: Asset price near $70
23 Asset Price at Maturity: $70.00
24 Payoff from Asset: $-0.72
25 Payoff from Call Options (Seller): $-0.00
26 Total P&L (Covered Call): $7191.24
27 P&L (Unhedged): $-0.72
28
29 Scenario: Asset price near $85
30 Asset Price at Maturity: $85.00
31 Payoff from Asset: $14995.35
32 Payoff from Call Options (Seller): $-9995.35
33 Total P&L (Covered Call): $12191.96
34 P&L (Unhedged): $14995.35
```

The result shows that, when the asset price is near \$70, which is below the strike price, the call options are not used. The investor retains the entire premium, resulting in a profit that comes solely from the initial option income. There is no asset-based gain or loss since the asset price ends where it started and no obligation to sell the shares arises. In this scenario, the covered call strategy outperforms simply holding the asset, as the unhedged position would show little to no profit, while the covered call earns the premium.

When the asset price ends near \$85, which is above the strike price, the call options are exercised. The investor must sell the shares at the strike price, capturing the difference between the strike and the initial price plus the premium received.

Although this scenario shows a substantial gain, it is less than what the unhedged position would have earned if the shares had been sold at a higher market price without option obligations. Thus, the covered call strategy provides additional income over small price increases, but sacrifices some of the upside potential, capping the maximum profit achievable.

5.3 Summary

In this chapter, we explored the foundational concepts and practical implementations of risk management techniques using financial derivatives. Beginning with an overview of the importance of hedging in uncertain and volatile markets, we highlighted the necessity for strategies that reduce unwanted exposure while still preserving the potential for gains where possible.

Futures contracts were introduced as a straightforward tool for hedging price risk. Their standardized nature and high liquidity make them suitable for mitigating uncertainty in commodities, currencies, and financial products. We discussed the mechanics of setting up a hedge with futures and derived the mathematical relationship between spot and futures positions. This included examining the optimal hedge ratio, which minimizes variance, and demonstrating the effectiveness of futures-based hedging through scenario analysis. Although such hedges can significantly reduce risk, we emphasize that risk and imperfect correlations remain persistent challenges, preventing complete elimination of uncertainty.

We then moved on to options-based hedging strategies, focusing on protective put and covered call strategies. By purchasing a protective put, an investor establishes a safety net against adverse price movements while still retaining the opportunity to benefit from price increases. In contrast, the covered call strategy allows an investor who already owns an asset to generate additional income from premiums, albeit at the cost of capping potential upside if prices rise substantially. Through several examples of scenario analysis, we show how these strategies strike different balances between risk reduction, cost, and forgone opportunities.

To provide a deeper understanding, we implemented these hedging strategies in Python. By simulating asset and futures price paths with Geometric Brownian Motion and pricing options using the Black-Scholes-Merton model, we illustrated how changes in underlying parameters, market conditions, and hedge ratios affect the outcomes of various strategies. Dynamic hedging approaches, which require continual monitoring and adjustment of hedge positions, were introduced to better reflect the evolving nature of real-world markets.

In summary, this chapter demonstrated that while no hedge is perfect, the informed use of futures and option contracts allows for more controlled exposure to market volatility. By thoughtfully selecting hedge ratios, strike prices, and maturities, and embracing dynamic adjustments in response to changing conditions, practitioners can significantly reduce risk and improve the predictability of portfolio performance. The techniques presented here form the foundation for more advanced risk management frameworks and guide the practical application of financial derivatives as essential instruments in modern finance.

Effective risk management is essential to preserve the stability and resilience of financial markets, institutions, and individual portfolios, as unmanaged exposures can lead to significant losses or systemic disruptions due to various types of risks. One prominent method to mitigate these risks is hedging with derivatives, as discussed in Chapter 5. This involves the systematic deployment of instruments such as options, futures, forwards, and swaps to compensate for adverse fluctuations in underlying asset prices, thus cushioning the negative impact. For example, Chapter 5 introduced the use of futures contracts to lock in future transaction prices and protective puts to protect against sudden declines in asset prices.

When we invest in options, another commonly used hedging technique is delta hedging, which is aimed at stabilizing the option price due to fluctuations in the price of the underlying asset. Options are powerful vehicles for both leverage and hedging, yet their prices often display high sensitivity to the respective underlying. The goal of delta hedging is thus to construct portfolios that are locally insensitive to small changes in the price of the underlying asset; that is, setting the derivative of the portfolio value with respect to the asset price (the delta) to zero. In other cases, hedging approaches may involve minimizing portfolio variance by solving optimization problems where the objective is to minimize the quadratic risk measure subject to the constraints imposed by the derivative contracts, thus stabilizing returns and securing the portfolios against unfavorable price movements.

In modern finance, derivatives are acknowledged as versatile instruments that serve both hedging and leveraging functions, and their efficacy can often be analyzed using mathematical frameworks. For example, options provide the holder with the right, though not the obligation, to buy or sell an underlying asset at a predetermined strike price before or on a specified expiration date. These instruments are often modeled using the Black-Scholes framework, which uses stochastic differential equations and assumes a geometric Brownian motion to derive closed-form pricing solutions. In contrast, futures contracts impose a compulsory obligation to transact at a future date, with their valuation frequently based on cost-of-

carry models that integrate factors such as interest rates and storage costs to enforce no-arbitrage conditions. Similarly, forward contracts extend this concept into over-the-counter agreements, allowing for customized contractual terms that accommodate unique risk profiles. In addition, swaps facilitate the exchange of cash flows or other financial variables—such as fixed vs. floating interest rates or different currencies—by applying fixed-for-floating exchange models that help mitigate specific exposures. All these vehicles can be further calibrated by formulating different optimization problems and using variance minimization techniques, aiming at effectively managing risk and optimizing returns.

Hedging is fundamentally a risk mitigation strategy that aims to eliminate or significantly reduce exposure to adverse market movements rather than to generate profit. Often, such risk mitigation comes at the cost of potential upside gain. One example is to take offsetting positions in derivative products, such as futures and options, thus neutralizing the portfolio's sensitivity to fluctuations in underlying asset prices. As discussed in the previous chapter, an investor with a substantial position in a particular asset can choose to purchase put options (called the protective put strategy), using models such as Black-Scholes to calculate the option delta and determine the optimal hedge ratio to offset potential declines in the asset value. Similarly, a corporation that expects to earn income in a foreign currency may use currency swaps to stabilize its cash flows and minimize volatility. All of these hedging techniques aim to protect against unforeseen market fluctuations to obtain more stable and predictable outcomes in both investment portfolios and corporate financial operations.

To further elaborate on the concept of delta hedging, consider an option priced at $V(S, t)$, where S denotes the underlying asset's price and t represents the current time, which also relates to the residual time to expiration. The option's delta, Δ , is then defined as $\Delta = \frac{\partial V}{\partial S}$, which is the first derivative (instantaneous rate of change) of V against S . In a delta-neutral hedging strategy, the objective is to construct a portfolio (an additional position on top of the current option) in which the aggregate delta will be zero, thereby neutralizing the sensitivity of the option price to small fluctuations in the underlying asset's price. To achieve this, we can adjust a position in the underlying asset that has a hedge ratio of $-\Delta$; that is, for every unit of delta exposure in the option, an opposing position of equal magnitude is taken in the underlying asset. Given that Δ is inherently a function of both the asset price S and time t (owing to the dynamic nature of the option's payoff structure and market volatility), continuous or frequent rebalancing is required to maintain the hedge. For example, if an investor holds a call option and the underlying stock currently trades at \$100 with $\Delta = 0.5$, the corresponding hedge would be to short 0.5 unit of shares per option. Should the stock price increase to \$101, and the resulting delta become 0.55, the hedge must be adjusted by shorting an additional 0.05 shares per option to restore delta neutrality. This example illustrates the dynamic and mathematically intensive nature of delta hedging, where the principles of differential calculus and continuous-time finance underpin the iterative process of risk mitigation in volatile markets.

Hedging strategies can be broadly classified into dynamic and static approaches, each displaying different methodologies, benefits, and inherent trade-offs. Dynamic hedging involves continuously or frequently rebalancing portfolio positions to stabilize portfolio value and adapt to changing market conditions, so that the portfolio is less sensitive to fluctuations in the price of the underlying asset. However, this method requires one to continuously monitor the underlying asset price and make a proper adjustment, which would incur a high transaction cost. In contrast, static hedging promotes the construction of the hedge at the inception of a strategy and maintaining such hedging position throughout its duration, thus simplifying portfolio management. However, such a static nature may potentially fail to cater to rapid market shifts.

In contrast, static hedging involves establishing a hedge at the beginning of a position that remains unchanged until maturity, thus preventing the need for continuous rebalancing of the hedging position. Instead, it relies on the upfront selection of financial instruments that are engineered to replicate the target payoff profile from the beginning, ensuring that the hedge remains relatively stable and demands minimal monitoring throughout its duration. This approach, while operationally simpler and less burdened by transaction costs associated with frequent adjustments, inherently sacrifices flexibility because it does not adjust to unforeseen market movements or shifts in underlying risk factors, thereby potentially exposing the portfolio to residual risks (the portion of risk that remains after implementing the hedge) that may deviate from the original expectations. For example, consider the static replication of a European put option using put-call parity: by combining a position in the underlying asset, a bond that replicates the risk-free component (often represented as Ke^{-rT} in the pricing formula), and a call option with the same strike and maturity, one can replicate the eventual payoff of the put option according to the central put-call parity relationship $P + S = C + Ke^{-rT}$. This mathematically elegant construction illustrates how a static hedge is designed to mirror the desired payoff without necessitating ongoing adjustments, yet it also underscores the inherent trade-off between the simplicity and operational efficiency of static replication and the potential exposure to market dynamics that a more flexible, dynamic strategy might mitigate.

Both dynamic and static hedging methods approach risk management in contrasting yet complementary ways, reflecting distinct trade-offs between flexibility and operational simplicity. Dynamic hedging, for example, provides fine-tuned adaptability by continuously rebalancing positions in response to evolving market conditions and adjusting for sensitivities encapsulated by the option Greeks, such as delta, gamma, and vega, thus allowing practitioners to mitigate risk exposures in real time; however, this precision comes at the cost of increased complexity and higher transaction costs associated with frequent trading. In contrast, static hedging simplifies the risk management process by constructing a replicating portfolio at the inception of the strategy, which is then held constant until maturity, effectively minimizing transaction costs and reducing the need for constant supervision, yet at the expense of limited responsiveness to unexpected market movements or volatility shifts. In this chapter, we delve into the nuances of these methodologies

by examining dynamic hedging techniques such as continuous rebalancing and the explicit use of the option Greeks to adjust for market risk. We will also explore static methods that involve the careful construction of replicating portfolios, thus providing a comprehensive framework that underscores both the theoretical foundations and practical implications of these complementary approaches to risk management.

Let us start with dynamic hedging.

6.1 Dynamic Hedging

Dynamic hedging focuses on continuously adapting a hedging position in response to evolving sensitivities of an option, collectively known as Greeks, that quantify the derivatives of the option price with respect to various underlying parameters. These parameters include Δ (delta), which measures the first-order sensitivity to changes in the underlying asset's price; Γ (gamma), representing the second-order sensitivity that captures the curvature of the price response; ν (often denoted as Vega), which gauges the sensitivity to changes in volatility; Θ (theta), reflecting the rate of time decay; and ρ (rho), quantifying the sensitivity to interest rate fluctuations. In practical applications, dynamic hedging typically emphasizes adjustments based on Δ and Γ , as these factors play an important role in determining the necessary modifications to the hedging when the price of the underlying asset changes. Careful rebalancing of hedging positions based on these Greeks can effectively mitigate undesired risk exposures that may arise from nonlinear effects or market volatility. However, achieving such precision comes at a cost: the hedging strategy requires frequent trading, which inherently increases transaction fees and operational expenses, in addition to frequent attention given to the hedging portfolio.

The Black-Scholes model provides much of the theoretical foundation for dynamic hedging by offering a rigorous mathematical framework derived from no-arbitrage principles and continuous-time stochastic processes. Central to this framework is the Black-Scholes partial differential equation (PDE):

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0,$$

where V denotes the option price, σ represents the volatility of the underlying asset, S is the asset's current price, and r is the risk-free interest rate. This PDE encapsulates the dynamic interplay between the stochastic evolution of the underlying asset and the deterministic drift provided by the risk-free rate under the risk-neutral measure, and its solution yields closed-form formulas for European call and put options. These explicit pricing formulas not only facilitate the valuation of options but also provide the basis for calculating the Greeks, which represent the sensitivities of the option price to changes in various parameters such as the underlying asset's price (Δ), its curvature (Γ), volatility (ν or Vega), time decay (Θ), and interest rates (ρ). Consequently, these sensitivities are instrumental in informing

dynamic hedging strategies, as they determine how the hedging portfolio should be continuously rebalanced in response to market movements, volatility shifts, and the passage of time.

The following section investigates deeper the details of the dynamic delta hedging strategy.

6.1.1 Dynamic Delta Hedging Strategy

As introduced in Chapter 5, the Black-Scholes model establishes a rigorous framework for option pricing by providing closed-form formulas for European options, where the call option price is given by

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

and the put option price by

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1),$$

with the variables d_1 and d_2 defined as

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

where S_0 is the current underlying price, K the strike price, r the risk-free interest rate, T the time to maturity, and σ the volatility. To further elaborate on these concepts, consider an example where $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 5\%$ (or 0.05), and $\sigma = 25\%$ (or 0.25), and assume that one standard option contract corresponds to 100 shares. Here, since $\ln(S_0/K) = \ln(1) = 0$, the numerator in the expression for d_1 simplifies to $0 + \left(0.05 + \frac{0.25^2}{2}\right)T = 0.05 + 0.03125 = 0.08125$; dividing by $\sigma\sqrt{T} = 0.25 \times 1$ yields $d_1 \approx \frac{0.08125}{0.25} = 0.325$, and hence $d_2 \approx 0.325 - 0.25 = 0.075$. Using standard normal tables or numerical routines, one obtains $N(d_1) \approx 0.628$ and $N(d_2) \approx 0.53$; furthermore, calculating $e^{-rT} \approx e^{-0.05} \approx 0.9512$ enables us to determine the first term of the call price as $S_0 N(d_1) = 100 \times 0.628 = 62.8$ and the second term as $K e^{-rT} N(d_2) = 100 \times 0.9512 \times 0.53 \approx 50.31$, thereby resulting in a per-share call price of approximately $C \approx 62.8 - 50.31 = 12.49$; when scaled to a standard contract of 100 shares, the call option is priced at roughly $12.49 \times 100 = 1249$.

Furthermore, the Black-Scholes delta for a call option, defined as $\Delta_{\text{call}} = N(d_1)$, is approximately 0.628 per share; therefore, for a contract representing 100 shares, the total delta is approximately 62.8, indicating that to achieve a delta neutral hedge, one would need to short close to 63 shares per call contract, thus offsetting the sensitivity of the call option value to small changes in the price of the underlying

asset. This means that the current movement in the hedging portfolio is to short 63 shares of stock for each call option contract. When we move to the next period, another rebalancing is often required due to changes in the asset price.

6.1.2 Continuous Rebalancing and Gamma Hedging

Suppose that shortly after the initial setup, the underlying stock price increases to 102 in the next period, while the time to maturity T remains effectively one year due to negligible elapsed time. Recomputing the Black-Scholes parameters with $S = 102$, we recalculate d_1 as

$$d_1 = \frac{\ln(102/100) + \left(0.05 + \frac{0.25^2}{2}\right) \times 1}{0.25} = \frac{\ln(1.02) + 0.08125}{0.25} \approx 0.4042,$$

where the term $\ln(1.02)$ captures the logarithmic return from the price increase and the additive term 0.08125 incorporates the effects of both the risk-free rate and the half-variance adjustment. Consequently, d_2 is updated as

$$d_2 = d_1 - 0.25 \approx 0.1542.$$

Using a standard normal cumulative distribution function, we find $N(0.4042) \approx 0.6577$, which implies that the call option's per-share delta, defined as $\Delta_{\text{call}} = N(d_1)$, is approximately 0.6577. Since each contract corresponds to 100 shares, the contract delta becomes 65.77. Recall that prior to the price change, the portfolio was hedged by shorting 63 shares, which was designed to effectively offset an earlier call delta of 62.8 per contract. Now, the updated net delta of the portfolio can be calculated as

$$\Delta_{\text{net}} = \Delta_{\text{call,new}} \times 100 - 63 = 65.77 - 63 = 2.77,$$

indicating a slight net long exposure to the underlying asset. Therefore, to re-establish a delta-neutral position, one would need to short an additional 2.77 shares. However, given that trading is carried out in whole shares, this is typically approximated by selling 3 shares, which adjusts the total delta to

$$65.77 - 66 \approx -0.23,$$

a value that is effectively zero for the most practical hedging purposes.

In this dynamic delta hedging process, we continuously update the hedge by buying or selling small increments of the underlying asset as its price changes, ensuring that the overall portfolio delta remains near zero. Specifically, if we denote the option's current delta by $\Delta_{\text{option}} = \frac{\partial V}{\partial S}$ and the delta contribution from the short position in the underlying by $\Delta_{\text{underlying}}$ (with a negative sign when short), the goal

is to maintain $\Delta_{\text{net}} = \Delta_{\text{option}} + \Delta_{\text{underlying}} \approx 0$. Because the option's delta is a function of the underlying asset price S , it changes as S fluctuates, thus necessitating periodic adjustments. The sensitivity of delta to changes in S is quantified by gamma, defined as

$$\Gamma = \frac{\partial^2 V}{\partial S^2},$$

which indicates how rapidly Δ_{option} changes with S . A high gamma implies that even a small change in S can lead to a significant shift in delta, thus increasing the frequency and magnitude of the required rebalancing. For example, if the delta of the option increases from Δ_{old} to Δ_{new} after a price change, the adjustment needed in the underlying position is approximately $\Delta_{\text{new}} - \Delta_{\text{old}}$ per share, scaled by the number of shares in the contract.

To further optimize hedging efficiency, we may also introduce a gamma hedging position by incorporating an additional derivative whose gamma, Γ_{hedge} , takes an opposite sign to that of the primary option, so that the net gamma $\Gamma_{\text{net}} = \Gamma_{\text{option}} + \Gamma_{\text{hedge}}$ is minimized. This reduction in gamma sensitivity can substantially lower the number of delta rebalancing trades required.

However, in practice, the theoretical benefits of such continuous rebalancing or gamma hedging must be balanced against transaction costs, market liquidity constraints, and the practicalities of trading in discrete units (often whole shares). Table 6-1 provides a detailed numerical illustration of these concepts by summarizing how positions, delta, and gamma adjust when the stock price changes from \$100 to \$102, including the additional trades required to restore delta neutrality. In summary, updating the hedge requires recalculating the delta of the option with each significant price movement and executing an offsetting trade in the underlying to maintain $\Delta_{\text{net}} \approx 0$. In addition, the greater the gamma, the faster Δ changes with S , and consequently, the more frequent the rebalancing becomes, highlighting the intricate interaction between delta and gamma in dynamic hedging strategies.

Table 6-1 Delta and gamma hedge after the stock rises to \$102, starting from an initial delta hedge of 63 short shares at \$100

Scenario	Position	Delta/Gamma
Initial hedge at \$100	Long 1 call option	$\Delta_{\text{option}} \approx 62.8, \Gamma_{\text{option}}$
	Short 63 shares	$\Delta_{\text{shares}} = -63, \Gamma_{\text{shares}} = 0$
	Net position	$\Delta_{\text{net}} \approx -0.2, \Gamma_{\text{net}} = \Gamma_{\text{option}}$
Price increases to \$102	Long 1 call option	$\Delta_{\text{option.new}} \approx 65.77, \Gamma_{\text{option}}$
	Additional short 3 shares	$\Delta_{\text{new shares}} = -3, \Gamma_{\text{shares}} = 0$
	Net position after adjustment	$\Delta_{\text{net.new}} \approx -0.23, \Gamma_{\text{net}} = \Gamma_{\text{option}}$
Gamma hedge	Add opposing gamma position	$\Gamma_{\text{hedge}} = -\Gamma_{\text{option}}$
	Net position with gamma hedge	$\Delta_{\text{net}} \approx 0, \Gamma_{\text{net}} = 0$

6.1.3 Dynamic Hedging in Action

In this section, we implement a dynamic hedging strategy using historical price data from Tesla (TSLA) and Walmart (WMT) that span January 1, 2024, to December 1, 2024, and we begin by applying the Black-Scholes model to compute the main option parameters such as volatility (σ), strike price (K), and time to maturity (T). Specifically, we estimate σ from the historical logarithmic returns of the underlying asset, and then, using these inputs, we calculate the option price V via the Black-Scholes formula, which in turn enables us to compute the delta $\Delta = \frac{\partial V}{\partial S}$ of the option. Recall that for a European call option, this delta is given by $\Delta = N(d_1)$,

where $d_1 = \frac{\ln(S_0/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$ and $N(\cdot)$ denotes the standard normal cumulative distribution function.

Based on the calculated delta, if holding a call option, we short approximately $\Delta \times (\text{contract size})$ shares (or purchase shares in the case of a put option) to initiate a delta-neutral position. As the underlying price evolves throughout the year, the computed delta adjusts accordingly, necessitating periodic recalibration of the hedge at discrete time intervals rather than continuous monitoring. Thus, this dynamic delta hedging process can mathematically ensure that the net portfolio sensitivity, given by $\Delta_{\text{net}} = \Delta_{\text{option}} + \Delta_{\text{underlying}}$, remains close to zero, thus mitigating the risk associated with small fluctuations in S . However, it is important to note that such continuous updating, while theoretically sound, must also account for practical considerations such as transaction costs and market liquidity constraints, which may impact the frequency and feasibility of rebalancing in a real-world trading environment.

Figure 6-1 shows the cumulative wealth curve for both stocks, suggesting that TSLA is much more volatile than WMT for this period and therefore requires more frequent hedging actions.

Recall that the dynamic nature of the hedge becomes particularly evident as the underlying stock price evolves over time, requiring a systematic and continuous recalibration of the hedging position at each discrete trading interval. Specifically, on each trading day t , we calculate the delta of the option $\Delta_t = \frac{\partial V(S_t, t)}{\partial S}$, where

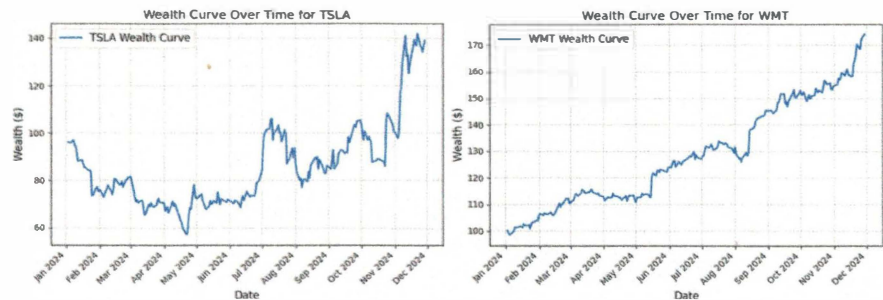


Figure 6-1 Cumulative wealth curve of both stocks

$V(S_t, t)$ represents the value of the option based on the current stock price S_t and the remaining time to maturity $T - t$. If this recalculated delta differs from the previous delta value Δ_{t-1} , the underlying position is adjusted by trading incremental $\Delta_t - \Delta_{t-1}$ shares, thus ensuring that the net portfolio delta, given by $\Delta_{\text{net}} = \Delta_t + \phi_t$ (with ϕ_t representing the current position in the underlying, typically negative for call options), remains approximately zero. This adjustment process can be formulated as maintaining $\Delta_t + \phi_t \approx 0$ at each rebalancing step. Importantly, transaction costs are incorporated into the model by associating a cost proportional to the absolute change in the underlying position, for example, $c \cdot |\Delta_t - \Delta_{t-1}|$, where c denotes the per-share transaction cost, thereby reflecting the financial impact of frequent rebalancing.

After each adjustment, the total portfolio value is updated to account for the current option price V_t , the market value of the underlying position $\phi_t S_t$, and the cash balance B_t , so that the overall portfolio value is given by $\Pi_t = V_t + \phi_t S_t + B_t$. The evolution of this portfolio value is then plotted to visually demonstrate how the dynamic hedge mitigates risk by stabilizing the portfolio's performance amid market fluctuations.

To further clarify the benefits of this strategy, we compare the performance of the dynamically hedged portfolio against that of an unhedged portfolio, which simply holds the option position without adjusting the underlying position. Performance metrics such as cumulative returns, computed as $R_{\text{cum}} = \prod_{t=1}^T (1 + r_t) - 1$, volatility σ measured by the standard deviation of returns, and the Sharpe ratio $SR = \frac{E[R - r_f]}{\sigma}$ (with r_f representing the risk-free rate) are calculated for both scenarios.

As shown in Figure 6-2, we can see how dynamic hedging often reduces volatility and improves risk-adjusted performance, even if it sometimes lowers absolute returns. This trade-off illustrates the essence of risk management: Although hedging may forgo some upside potential, it also protects against severe downside outcomes. This trade-off can also be formally expressed as a decrease in the tail risk, where the probability of extreme losses, measured by metrics such as Value at Risk (VaR) or Conditional Value at Risk (CVaR), is substantially lowered. Specifically, although such hedging may sacrifice some upside potential, since positive convexity benefits are partly dampened, it is a deliberate choice to mitigate severe downside outcomes.

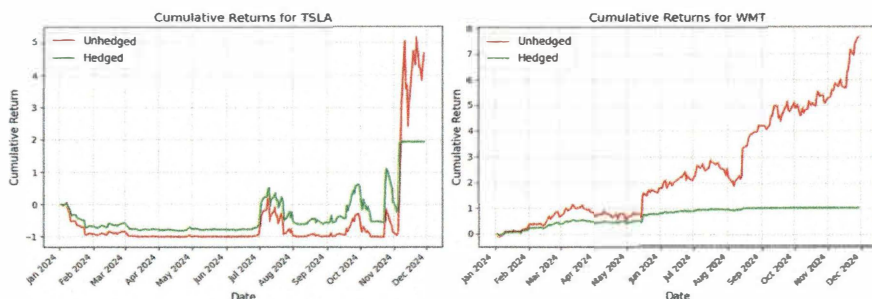


Figure 6-2 Comparing cumulative returns of both hedged and unhedged portfolios

Listing 6-1 implements this entire process step by step, performing a discrete approximation of continuously adjusting the hedge to keep the portfolio delta near zero. Note that for each ticker, we create a list S to store the cumulative return for each period to present the wealth curve, assuming a buy-and-hold strategy.

```
1 import numpy as np
2 import pandas as pd
3 import yfinance as yf
4 from scipy.stats import norm
5 import matplotlib.pyplot as plt
6 import warnings
7 import matplotlib.dates as mdates
8
9 warnings.filterwarnings('ignore')
10
11 # Parameters for the option
12 S0 = 100      # Initial stock price
13 K = 105       # Strike price
14 r = 0.05      # Risk-free interest rate
15 sigma = 0.2   # Volatility
16 T = 1         # Time to expiration in years
17
18 # Define tickers and date range
19 tickers = ['TSLA', 'WMT']
20 start_date = "2024-01-01"
21 end_date = "2024-12-01"
22
23 # Black-Scholes functions
24 def d1(S, K, r, sigma, T):
25     with np.errstate(divide='ignore', invalid='ignore'):
26         return (np.log(S / K) + (r + 0.5 * sigma ** 2) * T) / (
27             sigma * np.sqrt(T))
28
29 def delta_call(S, K, r, sigma, T):
30     d1_val = d1(S, K, r, sigma, T)
31     return norm.cdf(d1_val)
32
33 # Download and prepare data
34 stock_data = {}
35 stock_returns = {}
36 stock_S = {}
37
38 for ticker in tickers:
39     data = yf.download(ticker, start=start_date, end=end_date,
40                       interval='1d')
41     if data.empty:
42         raise ValueError(f"No data fetched for {ticker}. Check
43 the symbol or date range.")
44     close_prices = data['Close'].dropna()
45     returns = close_prices.pct_change().dropna()
46     S = [S0]
47     for ret in returns:
48         S.append(S[-1] * (1 + ret))
```

```

46 S = np.array(S)
47 stock_data[ticker] = close_prices
48 stock_returns[ticker] = returns
49 stock_S[ticker] = S
50
51 # Initialize portfolio metrics
52 portfolio_value = {ticker: [] for ticker in tickers}
53 deltas = {ticker: [] for ticker in tickers}
54 shares_held = {ticker: [] for ticker in tickers}
55 cash_position = {ticker: [] for ticker in tickers}
56
57 transaction_cost = 0.01
58
59 for ticker in tickers:
60     S_current = stock_S[ticker]
61     N = len(S_current)
62     dt = T / (N - 1)
63     t_min = 1 / 252
64
65     initial_delta = delta_call(S0, K, r, sigma, T)
66     initial_shares = initial_delta * 100
67     shares_held[ticker].append(-initial_shares)
68     cash = initial_delta * S0 * 100
69     cash_position[ticker].append(cash)
70     portfolio = initial_delta * S0 * 100 - cash
71     portfolio_value[ticker].append(portfolio)
72     deltas[ticker].append(initial_delta)
73
74     for i in range(1, N):
75         t = max(T - i * dt, t_min)
76         current_S = S_current[i]
77         current_delta = delta_call(current_S, K, r, sigma, t)
78         deltas[ticker].append(current_delta)
79
80         delta_change = (current_delta - deltas[ticker][i - 1]) *
81         100
82         shares_adjustment = np.round(delta_change)
83         shares_held[ticker].append(shares_held[ticker][-1] -
84         shares_adjustment)
85
86         cash_change = shares_adjustment * current_S * (1 +
87         transaction_cost)
88         cash_position[ticker].append(cash_position[ticker][-1] +
89         cash_change)
90
91         option_value = current_delta * current_S * 100
92         stock_position = shares_held[ticker][-1] * current_S
93         portfolio = option_value + stock_position + cash_position
94         [ticker][-1]
95         portfolio_value[ticker].append(portfolio)
96
97 # Plot wealth curves side by side
98 fig, axes = plt.subplots(1, 2, figsize=(14, 5))

```



```

94 for ax, ticker in zip(axes, tickers):
95     S = stock_S[ticker]
96     ax.plot(stock_data[ticker].index[1:], S[1:], label=f'{ticker}
97         Wealth Curve', linewidth=2)
98     ax.set_title(f'Wealth Curve Over Time for {ticker}', fontsize
99         =14)
100     ax.set_xlabel('Date', fontsize=12)
101     ax.set_ylabel('Wealth ($)', fontsize=12)
102     ax.grid(True, linestyle='--', alpha=0.7)
103     ax.legend(fontsize=12)
104
105     # Format the x-axis dates
106     import matplotlib.dates as mdates
107     ax.xaxis.set_major_locator(mdates.MonthLocator()) # Set
108     major ticks to monthly intervals
109     ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
110     # Format as 'Jan 2024', etc.
111     plt.setp(ax.get_xticklabels(), rotation=45, ha='right')
112
113 plt.tight_layout()
114 plt.show()
115
116 # Compare hedged vs unhedged portfolios
117 fig, axes = plt.subplots(1, 2, figsize=(14, 5))
118
119 for i, ticker in enumerate(tickers):
120     S = stock_S[ticker]
121     N = len(S)
122     dt = T / (N - 1)
123     t_min = 1 / 252
124     unhedged_portfolio = []
125     for j in range(1, N):
126         t = max(T - j * dt, t_min)
127         current_S = S[j]
128         d1_val = d1(current_S, K, r, sigma, t)
129         d2_val = d1_val - sigma * np.sqrt(t)
130         if np.isnan(d1_val) or np.isnan(d2_val) or np.isinf(
131             d1_val) or np.isinf(d2_val):
132             option_price = max(0, current_S - K) * 100
133         else:
134             option_price = (current_S * norm.cdf(d1_val) -
135                 K * np.exp(-r * t) * norm.cdf(d2_val)
136             ) * 100
137         unhedged_portfolio.append(option_price)
138
139     hedged_port = np.array(portfolio_value[ticker][1:])
140     unhedged_port = np.array(unhedged_portfolio)
141
142     unhedged_returns = (unhedged_port - unhedged_port[0]) /
143     unhedged_port[0]
144     hedged_returns = (hedged_port - hedged_port[0]) / hedged_port
145     [0]

```



```

139 axes[i].plot(stock_data[ticker].index[1:], unhedged_returns,
140 label='Unhedged', color='red')
141 axes[i].plot(stock_data[ticker].index[1:], hedged_returns,
142 label='Hedged', color='green')
143 axes[i].set_title(f'Cumulative Returns for {ticker}',
144 fontsize=14)
145 axes[i].set_xlabel('Date', fontsize=12)
146 axes[i].set_ylabel('Cumulative Return', fontsize=12)
147 axes[i].grid(True, linestyle='--', alpha=0.7)
148 axes[i].legend(fontsize=12)
149
150 axes[i].xaxis.set_major_locator(mdates.MonthLocator())
151 axes[i].xaxis.set_major_formatter(mdates.DateFormatter('%b %Y
152 '))
153 plt.setp(axes[i].get_xticklabels(), rotation=45, ha='right')
154
155 plt.tight_layout()
156 plt.show()
157
158 for ticker in tickers:
159     hedged_port = np.array(portfolio_value[ticker][1:])
160     S = stock_S[ticker]
161     N = len(S)
162     dt = T / (N - 1)
163     t_min = 1 / 252
164     unhedged_portfolio = []
165     for j in range(1, N):
166         t = max(T - j * dt, t_min)
167         current_S = S[j]
168         d1_val = d1(current_S, K, r, sigma, t)
169         d2_val = d1_val - sigma*np.sqrt(t)
170         if np.isnan(d1_val) or np.isnan(d2_val) or np.isinf(
171 d1_val) or np.isinf(d2_val):
172             option_price = max(0, current_S - K)*100
173         else:
174             option_price = (current_S*norm.cdf(d1_val) - K*np.exp
175 (-r*t)*norm.cdf(d2_val))*100
176         unhedged_portfolio.append(option_price)
177
178 unhedged_port = np.array(unhedged_portfolio)
179 unhedged_daily_returns = np.diff(unhedged_port) /
180 unhedged_port[:-1]
181 hedged_daily_returns = np.diff(hedged_port) / hedged_port
182[:-1]
183
184 def calculate_metrics(daily_returns):
185     cumulative_return = np.prod(1 + daily_returns) - 1
186     volatility = np.std(daily_returns)*np.sqrt(252)
187     sharpe_ratio = (np.mean(daily_returns) / np.std(
188 daily_returns))*np.sqrt(252)
189     return cumulative_return, volatility, sharpe_ratio
190
191

```

```

182 unhedged_cum_ret, unhedged_vol, unhedged_sharpe =
    calculate_metrics(unhedged_daily_returns)
183 hedged_cum_ret, hedged_vol, hedged_sharpe = calculate_metrics
    (hedged_daily_returns)
184
185 print(f"Risk Metrics for {ticker}:")
186 print("Unhedged: Cumulative Return = {:.4f}, Volatility =
    {:.4f}, Sharpe Ratio = {:.4f}".format(unhedged_cum_ret,
    unhedged_vol, unhedged_sharpe))
187 print("Hedged: Cumulative Return = {:.4f}, Volatility =
    {:.4f}, Sharpe Ratio = {:.4f}".format(hedged_cum_ret,
    hedged_vol, hedged_sharpe))
188 print()

```

Listing 6-1 Dynamic delta hedging for call option contract

The summary in Table 6-2 displays the overall performance outcomes achieved through dynamic hedging and quantitatively demonstrates how such strategies can modify a portfolio's risk and return profile. As illustrated in the table, while a hedged strategy may sometimes lead to lower absolute returns—implying that the expected growth rate $E[R]$ is reduced—the corresponding decrease in volatility σ tends to enhance the Sharpe ratio. For example, in the case of TSLA, the hedged portfolio exhibits a higher SR , indicating that despite sacrificing some of the upside potential, the reduction in risk more than compensates in terms of risk-adjusted returns. Conversely, for WMT, which shows a consistent upward trend during the analyzed period, the hedging mechanism dampens the opportunity for higher growth to some extent, as reflected in its lower Sharpe ratio. This trade-off, which is characterized by the interplay between expected return and volatility, highlights a central theme of risk management: accepting a modest reduction in $E[R]$ can lead to a significantly lower σ and thereby a more stable portfolio, ultimately improving the risk-adjusted return even if the absolute returns are somewhat diminished.

Now let us turn to the case of static hedging that involves a forward contract.

Table 6-2 Overall metrics for TSLA and WMT

Stock	Strategy	Cumulative return	Volatility	Sharpe ratio
TSLA	Unhedged	4.6767	851.1622	1.1297
TSLA	Hedged	1.9256	4.2331	1.5846
WMT	Unhedged	7.6567	0.9249	2.9973
WMT	Hedged	1.0247	0.3227	2.5643

6.2 Static Hedging

For certain derivatives, one can devise a hedge that requires minimal or no rebalancing over time—a technique referred to as static hedging. Mathematically, suppose the derivative has a payoff function $f(S_T)$ at expiration T , where S_T is the underlying asset's price at maturity. The objective of static hedging is to construct, at inception, a replicating portfolio comprising positions in readily available instruments with payoff functions $g_i(S_T)$ and weights ϕ_i such that

$$\sum_{i=1}^n \phi_i g_i(S_T) \approx f(S_T)$$

for all possible values of S_T . This equation, derived from no-arbitrage principles, is either solved analytically for simpler instruments or numerically for more complex structures. By establishing such a portfolio at the beginning, the hedge is constructed once and for all, remaining largely unchanged until maturity, thereby eliminating the need for continuous adjustments that are needed in dynamic hedging. In contrast, dynamic hedging involves the continuous recalibration of the hedge to maintain a risk-neutral stance, typically by adjusting the underlying asset position according to the option's delta $\Delta = \frac{\partial V}{\partial S}$ and monitoring higher-order sensitivities such as gamma $\Gamma = \frac{\partial^2 V}{\partial S^2}$ as the underlying price S_t evolves.

Static hedging, while more straightforward and often more cost-effective due to a lack of repeated transaction costs, may not be as responsive to unanticipated price movements. Nonetheless, when the replicating instruments are liquid and the derivative's payoff is amenable to such a precise static replication, this approach can efficiently approximate the target payoff and offer a robust risk management strategy without the cumulative errors and costs associated with continuous rebalancing.

Let us start by looking at statically hedging a forward contract.

6.2.1 Static Hedging for a Forward Contract

Determining the fair value of a forward contract involves setting a forward price K that precludes any arbitrage opportunities and aligns the contract's value with the cost of replicating its eventual payoff. By definition, a forward contract is an agreement made at inception (time $t = 0$) to transact an underlying asset at a predetermined price K at a future maturity date T . This means that, if priced correctly, the contract's initial value should be zero, thus ensuring fairness for both parties and eliminating the possibility of risk-free profits. To see this, consider an asset currently priced at $S_0 = 120$ with a proposed forward price $K = 125$, a continuously compounded risk-free rate $r = 0.03$ per annum, and a time to maturity $T = 2$ years. The underlying asset's dynamics under the real-world measure P can

be modeled by the stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t^P,$$

where μ is the drift and σ the volatility. However, in order to enforce no-arbitrage, we transition to the risk-neutral measure Q where the drift is replaced by the risk-free rate r , yielding

$$dS_t = r S_t dt + \sigma S_t dW_t^Q,$$

with W_t^Q representing a Brownian motion under Q . This change of measure guarantees that the discounted asset price $e^{-rt} S_t$ behaves as a martingale. Solving this SDE from time t to T results in

$$S_T = S_t \exp\left((r - \frac{1}{2}\sigma^2)(T - t) + \sigma(W_T^Q - W_t^Q)\right).$$

Taking the expectation under Q and applying the moment generating function of the normal distribution, $\mathbb{E}^Q\left[e^{\sigma(W_T^Q - W_t^Q)}\right] = e^{\frac{1}{2}\sigma^2(T-t)}$, we obtain

$$\mathbb{E}^Q[S_T | S_t] = S_t e^{r(T-t)}.$$

Thus, the time t value of the forward contract, which pays $S_T - K$ at maturity, is given by

$$\begin{aligned} F(S_t, t; K) &= e^{-r(T-t)} \mathbb{E}^Q[S_T - K | \mathcal{F}_t] = e^{-r(T-t)} (S_t e^{r(T-t)} - K) \\ &= S_t - K e^{-r(T-t)}. \end{aligned}$$

This formula establishes a clear relationship between the current asset price S_t , the forward price K (appropriately discounted to present value), and the contract's fair value. Notably, by choosing K such that $S_t = K e^{-r(T-t)}$, the forward contract is initially priced at zero, thereby preventing any immediate arbitrage opportunities. This static hedging approach, which replicates the derivative's payoff using a cost-of-carry argument, underscores the interplay between replication strategies and risk-neutral pricing in ensuring that forward contracts are fairly valued and that the market remains arbitrage-free.

To replicate the payoff of a forward contract using a static hedge, we begin by constructing a replicating portfolio at time $t = 0$ that requires no further adjustments until maturity. The strategy involves purchasing the underlying asset at its spot price S_0 and financing part of this acquisition by borrowing funds at the risk-free rate r . Specifically, suppose $S_0 = 120$ and we have a forward contract with a designated forward price K . At $t = 0$, you buy the asset for S_0 and simultaneously borrow an amount equal to the present value of K , which is calculated as $K e^{-rT}$ for a contract

with maturity T . For example, if $K = 125$, $r = 0.03$ per annum, and $T = 2$ years, then the amount borrowed is

$$Ke^{-rT} = 125e^{-0.03 \times 2} = 125e^{-0.06}.$$

Since $e^{-0.06} \approx 0.9418$, we have

$$125 \times 0.9418 \approx 117.72.$$

Thus, the net initial investment required to set up the hedge is

$$\text{Net Initial Investment} = S_0 - Ke^{-rT} = 120 - 117.72 = 2.28.$$

This net investment of 2.28 represents the initial value of the forward contract with the given parameters.

If our goal is to have a forward contract with zero initial cost—thereby eliminating any arbitrage opportunities—we set the forward contract's initial value to zero. Specifically, this is achieved by solving the equation

$$S_0 - Ke^{-rT} = 0,$$

which implies that

$$K = S_0 e^{rT}.$$

Substituting $S_0 = 120$, $r = 0.03$, and $T = 2$ yields

$$K = 120e^{0.06} \approx 120 \times 1.0618 \approx 127.42.$$

With $K \approx 127.42$, the forward contract is fairly priced with a zero initial net investment.

As time progresses toward maturity, the borrowed sum accrues interest deterministically at the rate r . Thus, by time T , the borrowed amount grows to

$$Ke^{-rT} e^{rT} = K.$$

At maturity, the asset is sold at the prevailing market price S_T , and the debt K is repaid. The net payoff from this replicating portfolio is then

$$\text{Net Payoff} = S_T - K,$$

which exactly mirrors the payoff of a long forward contract. For instance, if at maturity $S_T = 130$, the net payoff becomes $130 - 125 = 5$ (assuming the forward strike price is 125); alternatively, if $S_T = 115$, the net payoff is $115 - 125 = -10$.

This static replication is particularly efficient because it requires no rebalancing between $t = 0$ and $t = T$; the interest accrual is fully deterministic, and the asset does not generate intermediate cash flows (such as dividends) that might necessitate adjustments. Moreover, any deviation from the no-arbitrage condition (i.e., if the initial value of the replicating portfolio does not match the forward contract's value) would create an arbitrage opportunity, prompting market participants to exploit and thereby correct the mispricing.

To summarize the replication process, we have

- At $t = 0$: Purchase the underlying asset for S_0 and borrow Ke^{-rT} . The net cash outflow is

$$S_0 - Ke^{-rT}.$$

- Between $t = 0$ and $t = T$: Hold the asset without further adjustments while the borrowed amount accrues interest at the risk-free rate r , growing to K by time T .
- At $t = T$: Sell the asset for S_T and repay the debt of K . The net payoff is

$$S_T - K.$$

This one-time setup perfectly replicates the forward contract's payoff function, demonstrating that the forward's initial value must be consistent with the cost of establishing this static hedge.

Listing 6-2 illustrates this idea by plotting the net payoff $S_T - K$ for a range of possible maturity prices S_T . We set the initial parameters as above, simulate a range of values of S_T , and observe the linear relationship between S_T and the net payoff.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # Parameters
5 S0 = 120      # Initial asset price
6 K = 125       # Forward price
7 r = 0.03      # Risk-free interest rate (continuously compounded)
8 T = 2         # Time to expiration in years
9
10 # Present value of the forward price
11 PV_K = K * np.exp(-r * T)
12
13 # Initial net investment to replicate the forward
14 cost_asset = S0
15 borrow_amount = PV_K
16 net_investment = cost_asset - borrow_amount
17
18 print(f"Initial Investment to replicate forward: ${net_investment
19       :.2f}")
20
21 # Range of possible asset prices at maturity
```



```

21 S_T_values = np.linspace(80, 160, 500)
22
23 # Net payoff of the replication at maturity
24 net_payoff = S_T_values - K
25
26 # Plot the relationship
27 plt.figure(figsize=(10, 6))
28 plt.plot(S_T_values, net_payoff, color='blue')
29 plt.axhline(0, color='black', linewidth=0.5)
30 plt.title('Net Payoff of Static Hedge for a Forward Contract at
    Maturity')
31 plt.xlabel('Asset Price at Maturity ($)')
32 plt.ylabel('Net Payoff ($)')
33 plt.grid(True)
34 plt.show()
35
36 # Output
37 Initial Investment to replicate forward: $2.28

```

Listing 6-2 Static hedging for forward contract

As shown in Figure 6-3, the linearity visually confirms that the static hedge perfectly replicates the forward payoff and that the forward value is consistent with the non-arbitrage principles.

Therefore, by linking the forward contract to a straightforward buy-and-borrow strategy, we demonstrate mathematically that the no-arbitrage forward price must reflect both the time value of money and the carrying costs associated with holding the underlying asset. In this replication approach, an investor purchases the asset

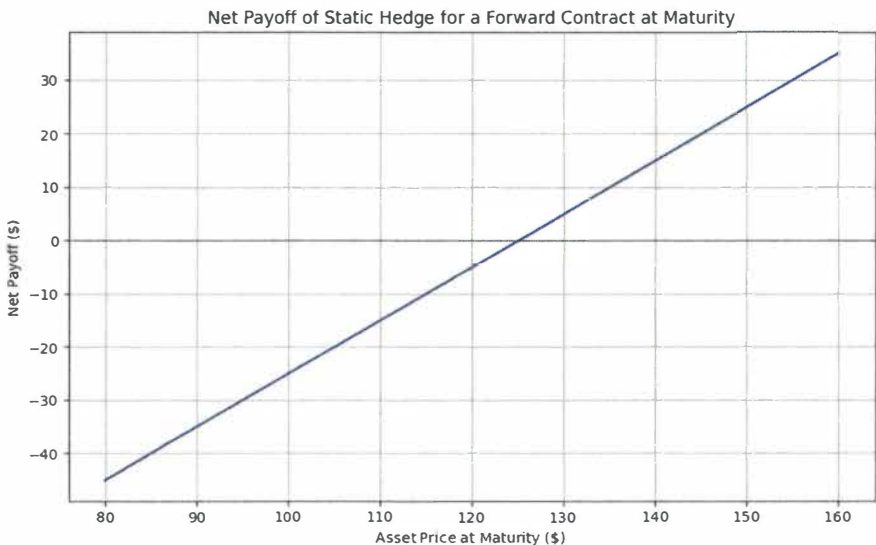


Figure 6-3 Net payoff curve of static hedge for a forward contract at maturity

at the current spot price S_0 and simultaneously borrows an amount equal to the present value of the forward price K , namely Ke^{-rT} , where r is the continuously compounded risk-free rate and T is the time to maturity. The net initial investment is thus

$$S_0 - Ke^{-rT}.$$

For the forward contract to be fairly priced—that is, to have zero initial value—this net investment must vanish, which requires that

$$S_0 - Ke^{-rT} = 0 \implies K = S_0e^{rT}.$$

This formula encapsulates the principle that the forward price grows at the risk-free rate, capturing the time value of money. In more general terms, if the underlying asset incurs additional carrying costs or provides a convenience yield, the forward price can be adjusted to

$$K = S_0e^{(r+u-c)T},$$

where u represents storage or other holding costs and c denotes any benefits such as a convenience yield. Any deviation from this fair pricing would yield a discrepancy between the cost of constructing the static replication portfolio and the contractual payoff $S_T - K$ at maturity, thereby opening arbitrage opportunities. Arbitrageurs would exploit such mispricing by, for instance, buying the asset and simultaneously selling the overpriced forward contract, or vice versa, until the no-arbitrage condition is restored. Thus, by ensuring that the initial value of the forward contract aligns precisely with the static replication cost, the forward price is set to prevent arbitrage and maintain market efficiency.

Next, we introduce static hedging for a European option.

6.2.2 Static Hedging for a European Put Option

To understand how to statically hedge a European put option, consider a put option with strike price K and maturity T , whose payoff at expiration is given by

$$(K - S_T)^+ = \max\{K - S_T, 0\},$$

where S_T is the price of the underlying asset at time T . This thresholding function implies that if $S_T < K$, the option pays $K - S_T$, and if $S_T \geq K$, the payoff is zero. The objective of a static hedge is thus to establish a replicating portfolio at the initial time t_0 that remains unchanged until maturity T and whose final payoff exactly matches that of the put option without requiring any rebalancing. To achieve this replication, we construct a portfolio comprising three components: first, a short position in one unit of the underlying asset, which yields a payoff of $-S_T$ at time

T ; second, a long position in K units of a zero-coupon bond maturing at T , each paying \$1, so that the bonds collectively yield K at maturity (with a present value of $Ke^{-r(T-t_0)}$ at time t_0 when r is the continuously compounded risk-free rate); and third, a long position in a European call option on the same underlying asset with the same strike K and maturity T , which at expiration pays $(S_T - K)^+$. Thus, the total payoff of the replicating portfolio at time T is given by

$$V_T = -S_T + K + (S_T - K)^+.$$

To verify that this portfolio indeed replicates the put payoff, we consider two cases. If $S_T \leq K$, the call option expires worthless so that $(S_T - K)^+ = 0$, and the portfolio payoff simplifies to

$$V_T = -S_T + K = K - S_T,$$

which is exactly the payoff $(K - S_T)^+$ since $K - S_T \geq 0$ in this region. Conversely, if $S_T > K$, the call option is in the money and pays $(S_T - K)^+ = S_T - K$, yielding

$$V_T = -S_T + K + (S_T - K) = 0.$$

In this scenario, the put option would also pay $(K - S_T)^+ = 0$ because $K - S_T < 0$, thereby ensuring that the portfolio payoff matches the put payoff in all cases.

This construction is intimately related to the concept of put-call parity, which for European options on non-dividend-paying stocks states that

$$C - P = S_0 - Ke^{-rT},$$

where C and P are the call and put prices, respectively, and S_0 is the current spot price of the underlying asset. Rearranging gives

$$P = C - S_0 + Ke^{-rT},$$

which is consistent with our replicating strategy: by taking a long call, shorting the underlying asset, and holding Ke^{-rT} in risk-free bonds, one can synthetically construct the payoff of a put option.

By strategically selecting these initial positions at t_0 , we create a portfolio that yields exactly

$$V_T = (K - S_T)^+$$

at maturity, without any subsequent rebalancing. The absence of rebalancing is a direct consequence of the deterministic nature of the bond's accrual and the fact that the underlying asset and call option positions do not require interim adjustments when their payoffs are already fixed by expiration. This static hedging approach not only provides a clear illustration of the relationship between put and call options,

the underlying asset, and the time value of money through discounting, but it also reinforces the fundamental principles of derivative pricing and risk management by demonstrating how arbitrage-free conditions can be maintained through perfect replication.

6.2.2.1 Put-Call Parity

The static hedging strategy for a European put option is fundamentally rooted in the put-call parity theorem, a cornerstone of options pricing for European options on non-dividend-paying assets. This theorem establishes that the sum of the put option price $P(S_t, t)$ and the current underlying asset price S_t is exactly equal to the sum of the call option price $C(S_t, t)$ and the present value of the strike price K , discounted over the time to maturity $T - t$ at the continuously compounded risk-free rate r . Mathematically, this relationship is expressed as

$$P(S_t, t) + S_t = C(S_t, t) + Ke^{-r(T-t)}.$$

Rearranging the equation to solve for the put price yields

$$P(S_t, t) = C(S_t, t) + Ke^{-r(T-t)} - S_t.$$

This expression shows that the price of a European put option can be replicated by constructing a portfolio comprising three elements:

1. A long position in one European call option with strike K and maturity T , which grants the right to purchase the underlying asset at K
2. A long position in a portfolio of zero-coupon bonds with a combined face value of K , whose present cost is $Ke^{-r(T-t)}$ (reflecting the time value of money)
3. A short position in one unit of the underlying asset, incurring a payoff of $-S_T$ at maturity

At expiration, the European call option pays $(S_T - K)^+$, the zero-coupon bonds mature to pay K , and the short asset position delivers $-S_T$, so that the total payoff of the replicating portfolio is

$$V_T = (S_T - K)^+ + K - S_T.$$

We can then consider the two cases (i.e., $S_T \leq K$ and $S_T > K$) to confirm that this portfolio replicates the put payoff $(K - S_T)^+$. Thus, by carefully selecting these initial positions, the replicating portfolio produces a payoff identical to that of the put option without requiring any dynamic rebalancing. This static hedge not only underscores the elegance of the put-call parity but also reinforces the consistency of option pricing. If market prices were to deviate from this parity, arbitrageurs could construct risk-free profit strategies by buying the underpriced instrument and selling the overpriced one until the equilibrium is restored.

Moreover, the put-call parity relationship can be extended to more complex scenarios. For instance, if the underlying asset pays a continuous dividend yield q , the modified put-call parity becomes

$$P(S_t, t) + S_t e^{-q(T-t)} = C(S_t, t) + K e^{-r(T-t)},$$

ensuring that even with dividend payments, the replicating portfolio remains aligned with no-arbitrage principles.

In summary, by linking the static hedging strategy for a European put option to the mathematical framework provided by put-call parity, we see that the correct pricing of options inherently accounts for both the time value of money and the cost (or benefit) of carrying the underlying asset. This framework guarantees that the cost of establishing the replicating portfolio, comprised of a call option, zero-coupon bonds, and a short position in the asset, exactly equals the price of the put option, thereby ensuring consistent, arbitrage-free pricing in efficient markets.

6.2.2.2 Derivation Using Risk-Neutral Valuation

To derive the relationship between the prices of European put and call options, we begin within the risk-neutral valuation framework. Under the risk-neutral measure Q , the present value at time t of a European put option with strike K and maturity T that pays $(K - S_T)^+$ is given by

$$P(S_t, t) = e^{-r(T-t)} \mathbb{E}^Q[(K - S_T)^+ | \mathcal{F}_t],$$

where r is the continuously compounded risk-free rate, and $\mathcal{F}_t$ represents the information available at time t . Similarly, the price of a European call option with the same strike and maturity, which pays $(S_T - K)^+$ at expiration, is expressed as

$$C(S_t, t) = e^{-r(T-t)} \mathbb{E}^Q[(S_T - K)^+ | \mathcal{F}_t].$$

A key step in establishing put-call parity is to consider the linear combination

$$P(S_t, t) + S_t - K e^{-r(T-t)}.$$

Substituting the risk-neutral valuation of the put option, we have

$$P(S_t, t) + S_t - K e^{-r(T-t)} = e^{-r(T-t)} \mathbb{E}^Q[(K - S_T)^+ | \mathcal{F}_t] + S_t - K e^{-r(T-t)}.$$

Notice that we can rewrite the current price S_t in its forward form by recalling that under the risk-neutral measure

$$\mathbb{E}^Q[S_T | \mathcal{F}_t] = S_t e^{r(T-t)}.$$

Thus, expressing S_t as $e^{-r(T-t)} \cdot S_t e^{r(T-t)}$, the equation becomes

$$P(S_t, t) + S_t - K e^{-r(T-t)} = e^{-r(T-t)} \left(\mathbb{E}^Q[(K - S_T)^+ | \mathcal{F}_t] + S_t e^{r(T-t)} - K \right).$$

Recognize that the term $S_t e^{r(T-t)} - K$ is equivalent to $\mathbb{E}^Q[S_T - K | \mathcal{F}_t]$. To combine the two components inside the expectation, consider the random variable

$$X = (K - S_T)^+ + (S_T - K).$$

We examine X under two cases. When $S_T \leq K$, we have $(K - S_T)^+ = K - S_T$ and $(S_T - K) = -(K - S_T)$, so that

$$X = (K - S_T) + (S_T - K) = 0.$$

When $S_T > K$, we have $(K - S_T)^+ = 0$ and $(S_T - K) = S_T - K$, yielding

$$X = 0 + (S_T - K) = S_T - K.$$

Thus, in all cases, we have

$$(K - S_T)^+ + (S_T - K) = (S_T - K)^+.$$

Substituting back, we obtain

$$P(S_t, t) + S_t - K e^{-r(T-t)} = e^{-r(T-t)} \mathbb{E}^Q[(S_T - K)^+ | \mathcal{F}_t] = C(S_t, t).$$

Rearranging the expression leads to the celebrated put-call parity formula:

$$P(S_t, t) = C(S_t, t) + K e^{-r(T-t)} - S_t.$$

This relationship shows that the price of a European put option can be directly determined from the price of a European call option, the current spot price S_t , and the present value of the strike price K . The derivation underscores how risk-neutral valuation and the no-arbitrage principle work together: if the put-call parity did not hold, market participants could construct arbitrage strategies to exploit the mispricing, thereby restoring equilibrium. This fundamental connection not only ensures consistency in option pricing but also serves as a critical tool for detecting mispricings and maintaining efficient market conditions.

6.2.2.3 Static Hedging of European Put Option in Action

To demonstrate the practical implementation of a static hedge for a European put option, we leverage put-call parity that specifies the mathematical link between the prices of European put and call options on the same underlying asset with identical

strike K and maturity T . Specifically, recall that we can write the put option price as

$$P(S_t, t) = C(S_t, t) + Ke^{-r(T-t)} - S_t.$$

This equation implies that by knowing the price of a call option, the current price S_t , and the discounted strike $Ke^{-r(T-t)}$, we can construct a replicating portfolio that perfectly mimics the payoff of the put option at expiration. In practical terms, to replicate the put payoff $(K - S_T)^+$ at maturity, one can establish a portfolio at $t = 0$ comprising three components: a long position in one European call option, a long position in zero-coupon bonds with a face value of K (which cost Ke^{-rT} at inception), and a short position in one unit of the underlying asset. At maturity, the call option yields $(S_T - K)^+$, the bonds mature to pay K , and the short asset position results in a payoff of $-S_T$; together, these deliver a total payoff of

$$(S_T - K)^+ + K - S_T,$$

which simplifies to $(K - S_T)^+$ when considering the two cases $S_T \leq K$ and $S_T > K$.

In the code provided below, we first calculate the price of the European call option using the Black-Scholes formula, which requires as inputs the current asset price S_0 , strike K , time to maturity T , risk-free rate r , and volatility σ . For our demonstration, we set the parameters as follows: $S_0 = 120$, $K = 110$, $r = 0.03$ (3% per annum), $T = 2$ years, and $\sigma = 0.25$ (25% annual volatility). With these inputs, the Black-Scholes model computes the call price $C(S_0, 0)$; then, using the put-call parity relation, we derive the corresponding put price via

$$P(S_0, 0) = C(S_0, 0) + Ke^{-rT} - S_0.$$

Subsequently, we simulate a range of possible terminal asset prices S_T and calculate the final payoffs for both the replicating portfolio and the European put option. These simulations serve to validate that the static hedge, established at inception without any need for rebalancing, indeed produces a terminal payoff identical to $(K - S_T)^+$. As illustrated in Listing 6-3 and Figure 6-4, the payoff profiles of the replicating portfolio and the put option are virtually indistinguishable, confirming the effectiveness of the static hedging strategy. This exercise not only provides a practical demonstration of put-call parity in action but also reinforces the theoretical underpinnings of static hedging and no-arbitrage pricing in the options market.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import norm
4
5 def black_scholes_call(S, K, T, r, sigma):
6     """
7     Compute the Black-Scholes price for a European call option.
8     S: current asset price
9     K: strike price
10    T: time to maturity in years
11    r: risk-free interest rate (annual, continuously compounded)
12    sigma: volatility (annual)
13    """
14    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np
15    .sqrt(T))
16    d2 = d1 - sigma * np.sqrt(T)
17    call = S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
18    return call
19
20 # Parameters
21 S0 = 120          # Current underlying asset price
22 K = 110           # Strike price
23 r = 0.03          # Risk-free rate (3%)
24 T = 2             # Time to maturity (2 years)
25 sigma = 0.25      # Volatility (25%)
26
27 # Compute the call option price using Black-Scholes
28 call_price = black_scholes_call(S0, K, T, r, sigma)
29
30 # Compute the present value of the strike price
31 PV_K = K * np.exp(-r * T)
32
33 # Put-call parity:  $P = C + PV(K) - S_0$ 
34 put_price_implied = call_price + PV_K - S0
35
36 print(f"European Call Option Price: ${call_price:.2f}")
37 print(f"Present Value of Strike: ${PV_K:.2f}")
38 print(f"Put Price from Put-Call Parity: ${put_price_implied:.2f}"
39       )
40
41 # Simulate possible asset prices at maturity
42 S_T_values = np.linspace(60, 180, 500)
43
44 # Compute the payoff of the replicating portfolio
45 # Replicating portfolio = short underlying + K zero-coupon bonds
46 # + long call
47 payoff_short_asset = -S_T_values
48 payoff_bonds = K
49 payoff_call = np.maximum(S_T_values - K, 0)
50 portfolio_payoff = payoff_short_asset + payoff_bonds +
51     payoff_call
52
53 # Compute the actual put payoff for comparison

```

```

50 put_payoff = np.maximum(K - S_T_values, 0)
51
52 # Plot the payoffs to compare
53 plt.figure(figsize=(10, 6))
54 plt.plot(S_T_values, portfolio_payoff, label='Replicating
    Portfolio Payoff', color='blue')
55 plt.plot(S_T_values, put_payoff, label='European Put Option
    Payoff', color='red', linestyle='--')
56 plt.axhline(0, color='black', linewidth=0.5)
57 plt.title('Static Hedge Replicating European Put Option Payoff')
58 plt.xlabel('Asset Price at Maturity ($)')
59 plt.ylabel('Payoff ($)')
60 plt.legend()
61 plt.grid(True)
62 plt.show()
63
64 # Outputs of interest
65 print(f"With these parameters, the static hedge set up at time 0
    costs: ${put_price_implied:.2f}")
66
67 # Ouputput
68 European Call Option Price: $25.21
69 Present Value of Strike: $103.59
70 Put Price from Put-Call Parity: $8.80
71 With these parameters, the static hedge set up at time 0 costs: \
    $8.80

```

Listing 6-3 Static hedging for replicating European put option payoff function

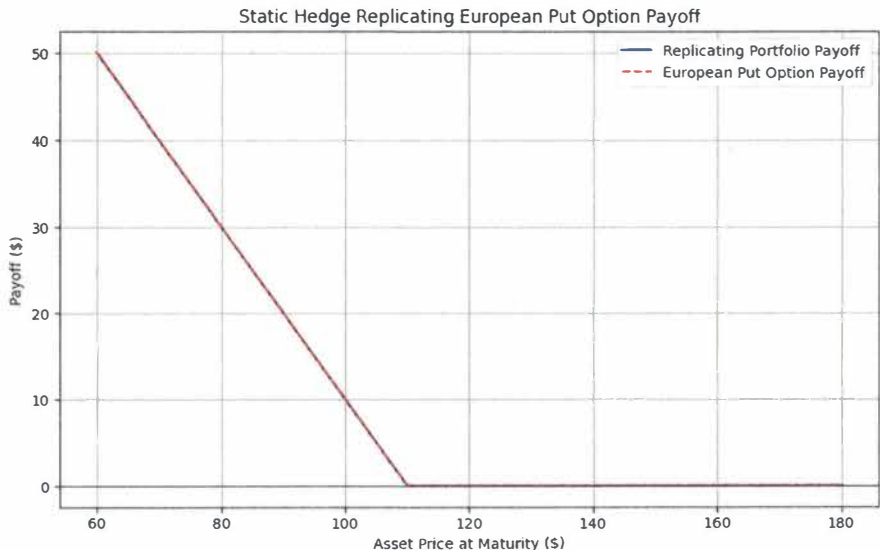


Figure 6-4 Replicating European put option via put-call parity